

Introduction to Digital Communications
(Course Reader 2003-2004)

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Chapter 1

Introduction and background

Modern telecommunication systems are formed by terminals exchanging *information* by transmitting *signals* through some transmission resource called *channel*. The ensemble of rules allowing this exchange of information is called *protocol*. The ensemble of terminals, channels and protocols forms a *telecommunication network*.

Protocols have been organized by the International Standard Organization into a reference hierarchical system, called Open System Interconnection (ISO-OSI). The goal of the ISO-OSI reference model is to provide a layered structure where upper layers can be built on top of existing lower layers without redesigning the whole network from scratch, and where existing higher levels can use new lower layers in a transparent way. The ISO-OSI is composed by 7 layers: Physical, Data Control Link, Network, Transport, Session, Presentation, Application. In this course, we are concerned with the Physical Layer. Moreover, we restrict our study to the simple case of one-way point-to-point transmission.

1.1 Reference model for one-way point-to-point transmission

We consider the simple idealized one-way point-to-point communication system represented in Fig. 1.1. This is composed by: information source, source encoder, channel encoder, signal modulator, physical channel, signal demodulator, channel decoder, source decoder, end user.

Information source. What is information? Intuitively, an event brings us some information if its occurrence is unexpected. In 1948, Claude Shannon defined in a mathematically rigorous way the theory of information *contained* in random variables and processes, and the problem of its reliable transmission over communication channels [4]. His seminal work is now considered as the birth of *Information Theory*, and more in general of modern communication theory [1].

Without getting into the details, we can say that an information source (e.g., speech, video) is characterized by its *rate-distortion function* $R(D)$, that gives the theoretical minimum number of bit/s (rate) necessary to represent the source with distortion D , according to some distortion measure. In the case of discrete sources (e.g., a data file), the minimum rate necessary to represent the source with zero distortion is called *entropy rate*.

Source encoder. The task of the source encoder is to translate the source signal into a sequence of bits at rate $R_b \geq R(D)$, for a given desired distortion level D . Examples of source

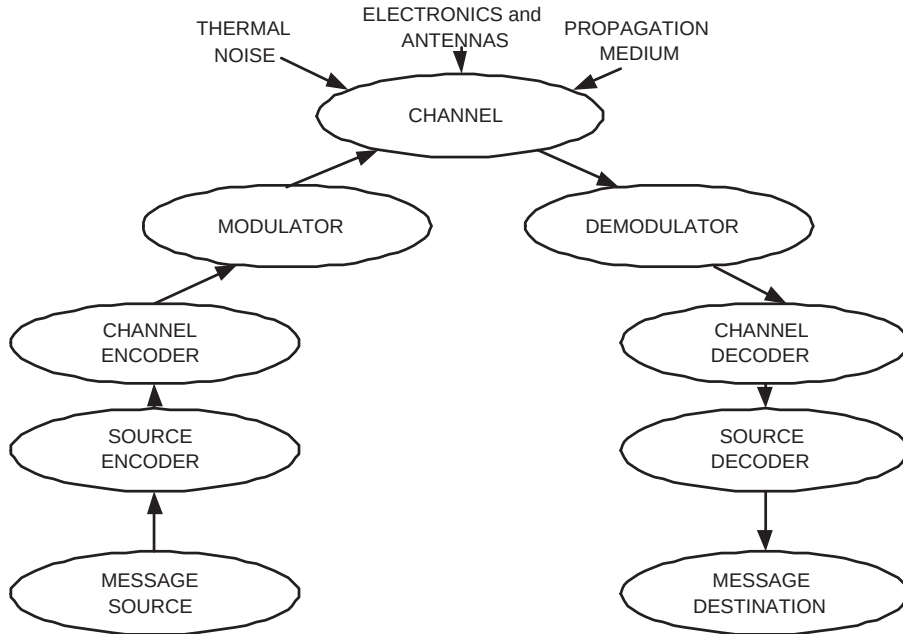


Figure 1.1: Block diagram of a one-way point-to-point communication system

encoders for analog sources are: PCM quantizers, vector quantizers, vocoders [3], still images and video compression schemes (e.g., JPEG, MPEG), subband coding schemes for audio. Examples of zero-distortion discrete source encoders are: Huffman encoders, Lempel-Ziv encoders [1] and various practical file compression schemes.

Channel encoder. The channel encoder takes the sequence of bits produced by the source encoder (ideally i.i.d.) and map them onto a sequence of channel symbols belonging to a given channel input alphabet $\mathcal{X}$. The set of all possible code sequences $\mathcal{C} \subseteq \mathcal{X}^N$ (where N denotes the sequence length) is called *codebook*, or simply *code*. Each code sequence $\mathbf{x} \in \mathcal{C}$ is called *codeword*. The code is constructed in order to “protect” the input information sequence from possible transmission errors occurring on the channel.

The code rate, expressed in bit per channel symbol, is given by $R = K/N$, where K is the length of the input information sequence (in bits). The symbol rate, i.e., the number of symbols per second, necessary to transmit the bit rate R_b is given by $R_s = R_b/R$.

Perhaps the main result of Shannon Theory is that nearly every physically meaningful communication channel is characterized by a rate C bit/s, called *channel capacity*, such that if $R_b < C$, there exist channel codes with arbitrarily small error probability, while if $R_b > C$, there are no such codes, and in many cases of interest the probability of error is close to 1.

A source is said to be transmissible with distortion D over a channel with capacity C if $R(D) < C$.

Signal modulator. The task of the signal modulator is to turn the codeword that must be transmitted over the channel into a signal able to travel over the physical channel without too

much distortion. Three examples are given in Figure 1.2. The first is was used in 300 bit/s (Baud) telephone MODEMs and is an example of *Binary Frequency Shift Keying (FSK)*, which we will revisit later in the course. The signals conveying 0 and 1 are two sinusoids of different frequency. The second example is used in T1 transmission systems for transmitting 1.5 Mbit/s over a copper wire. It is based on *Alternate Mark Inversion*, where a 1 is transmitted by a positive or negative polarity depending on whether the previous bit was a 1. Adjacent 1's use opposite polarity in order to avoid a DC component in the transmitted signal. 0's are conveyed by not transmitting any signal. The third example is a *Quarternary Pulse-Position Modulation* which is common in optical transmission. Here two bits are conveyed every 25 ns by modulating the position (among 4) of a pulse in the interval of 25ns.

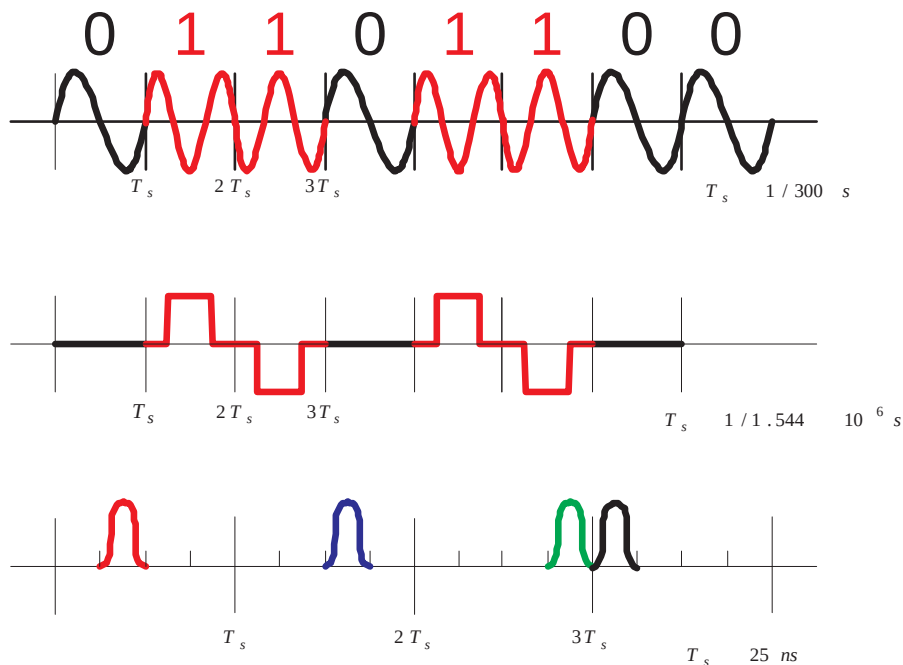


Figure 1.2: Three examples of Digital Modulation

Physical channel. We include as part of the physical channel all electronic devices (filters, amplifiers, antennas, free space propagation, cables, wires) and all external disturbances (thermal noise, interference) that may distort the signal produced by the modulator.

We restrict our treatment to linear time-invariant (LTI) additive-noise channels, whose relationship between the input signal $x(t)$ and the output signal $y(t)$ is given by

$$y(t) = h(t) \otimes x(t) + n(t) \quad (1.1)$$

where $h(t)$ is the channel impulse response, $n(t)$ is additive noise and $\otimes$ denotes convolution.

Demodulator. The task of the demodulator is to process the received signal output by the channel and to produce an *observation* from which the channel decoder can recover the transmitted codeword.

Channel decoder. The channel decoder decides (or *detects*) which codeword was transmitted, based on the observation of the received signal produced by the demodulator. The detected codeword is then mapped back to the corresponding information bit sequence.

Source decoder. The source decoder reproduces the original source signal from the bit sequence recovered by the channel decoder. If the sequence does not contain errors, it is able to represent the source with the desired level of distortion.

1.2 Signals and systems

In this section we recall some basic facts about deterministic and stochastic signals and systems.

1.2.1 Finite-energy and finite-power deterministic signals

The signal $x(t)$ is said to have finite energy if

$$\mathcal{E}_x \triangleq \int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$$

and it is said to have finite power if

$$\mathcal{P}_x \triangleq \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt < \infty$$

For signals of the first type, we define the autocorrelation function of $x(t)$ as

$$\phi_x(\tau) \triangleq \int_{-\infty}^{\infty} x(t)x(t-\tau)^* dt$$

For signals of the second type, we define the time-averaged autocorrelation function

$$\phi_x(\tau) \triangleq \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau)^* dt$$

Let $\mathcal{F}[\cdot]$ denote the Fourier transform operator, such that $X(f) = \mathcal{F}[x] = \int x(t)e^{-j2\pi ft} dt$. For a finite-energy signals $x(t)$, $|X(f)|^2 = \mathcal{F}[\phi_x]$ is called *energy spectral density* (ESD). In fact, because of Parseval identity,

$$\int_{-\infty}^{\infty} |X(f)|^2 df = \phi_x(0) = \mathcal{E}_x$$

For a finite-power signals $x(t)$, we define the *power spectral density* (PSD) $S_x(f) \triangleq \mathcal{F}[\phi_x]$. In fact, $\int_{-\infty}^{\infty} S_x(f) df = \phi_x(0) = \mathcal{P}_x$.

The output of a LTI system with impulse response $h(t)$ to the input $x(t)$ is given by the convolution integral

$$y(t) = h(t) \otimes x(t) \triangleq \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau$$

In the frequency domain, we have $Y(f) = H(f)X(f)$, where $H(f) = \mathcal{F}[h]$ is the system *transfer function*. The ESD (resp. PSD) of $y(t)$ and $x(t)$ are related by: $|Y(f)|^2 = |H(f)|^2 |X(f)|^2$ (resp. $S_y(f) = |H(f)|^2 S_x(f)$), where $|H(f)|^2$ is the system energy (resp. power) transfer function. In the time domain, we have

$$\phi_y(\tau) = \phi_h(\tau) \otimes \phi_x(\tau)$$

1.2.2 Random processes

Roughly speaking, a random process $x(t)$ can be seen either as a sequence of random variables $x(t_1), x(t_2), \dots, x(t_n)$, indexed by the “time” index $t = t_1, t_2, \dots$, or as a collection of signals $x(t; \omega)$, where ω is a random experiment taking on values in a certain event space Ω . The full statistical characterization of a random process $x(t)$ is given by the collection of all joint probability cumulative distribution function (cdf)

$$\Pr(x(t_1) \leq x_1, x(t_2) \leq x_2, \dots, x(t_n) \leq x_n)$$

for all $n = 1, 2, \dots$ and for all instants $t_1, t_2, \dots, t_n$.

Complex processes and random variables are characterized by the joint statistics of its real and imaginary parts. For example, a random variable $X = X_1 + jX_2$ is characterized by the joint cdf $\Pr(X_1 \leq x_1, X_2 \leq x_2)$.

The first and second order statistics of $x(t)$ are defined by its mean

$$\mu_x(t) = E[x(t)]$$

and by its autocorrelation function

$$\phi_x(t_1, t_2) = E[x(t_1)x(t_2)^*]$$

The average energy content of a random process $x(t)$ is given by

$$\begin{aligned} \mathcal{E}_x &= E \left[\int_{-\infty}^{\infty} |x(t)|^2 dt \right] \\ &= \int_{-\infty}^{\infty} \phi_x(t, t) dt \end{aligned}$$

The average power content of a random process $x(t)$ is given by

$$\begin{aligned} \mathcal{P}_x &= E \left[\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \right] \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \phi_x(t, t) dt \end{aligned}$$

The PSD of a random process $x(t)$ is defined in general by

$$S_x(f) \triangleq E \left[\lim_{T \rightarrow \infty} \frac{|X_T(f)|^2}{T} \right]$$

where $X_T(f) = \int_{-T/2}^{T/2} x(t) e^{-j2\pi ft} dt$. We have the following result [3]:

Wiener-Khinchin theorem. If for all finite τ and any interval $\mathcal{J}$ of length $|\tau|$ the autocorrelation function of the random process $x(t)$ satisfies the condition

$$\left| \int_{\mathcal{J}} \phi_x(t + \tau, t) dt \right| \leq \infty$$

the PSD of $x(t)$ is the Fourier transform of $\bar{\phi}_x(\tau)$, given by

$$\bar{\phi}_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \phi_x(t + \tau, t) dt$$

□

For two random processes $x(t)$ and $y(t)$, defined on a joint probability space, we define the *cross-correlation* function

$$\phi_{xy}(t_1, t_2) = E[x(t_1)y(t_2)^*]$$

and the cross-spectrum

$$S_{xy}(f) \triangleq E \left[\lim_{T \rightarrow \infty} \frac{X_T(f)Y_T(f)^*}{T} \right]$$

WSS and WSC processes. A random process $x(t)$ is said to be wide-sense stationary (WSS) if

1. $\mu_x(t) = \mu_x$ is constant with t .
2. $\phi_x(t_1, t_2)$ depends only on the difference $\tau = t_1 - t_2$ (for brevity, we shall use the notation $\phi_x(\tau) \triangleq \phi_x(t + \tau, t)$).

Two random processes $x(t)$ and $y(t)$ are said to be jointly WSS if both $x(t)$ and $y(t)$ are individually WSS and if their cross-correlation function $\phi_{xy}(t_1, t_2)$ depends only on the difference $t_1 - t_2$ (again, we shall use the notation $\phi_{xy}(\tau)$ in this case).

A random process $x(t)$ is said to be wide-sense cyclostationary (WSC) if

1. $\mu_x(t)$ is periodic of period T , i.e., $\mu_x(t) = \mu_x(t + kT)$, for $k \in \mathbb{Z}$.
2. $\phi_x(t_1, t_2)$ is periodic of period T , i.e., $\phi_x(t_1, t_2) = \phi_x(t_1 + kT, t_2 + kT)$, for $k \in \mathbb{Z}$.

For WSS processes, we have that the Wiener-Khinchin theorem holds and yields

$$S_x(f) = \mathcal{F}[\phi_x] = \int_{-\infty}^{\infty} \phi_x(\tau) e^{-j2\pi f\tau} d\tau$$

For WSC processes, if $|\int_0^T \phi_x(t + \tau, t) dt| < \infty$, then the Wiener-Khinchin theorem yields

$$S_x(f) = \mathcal{F}[\bar{\phi}_x]$$

where $\bar{\phi}_x(\tau)$ is obtained by time-averaging the autocorrelation function over one period, i.e.,

$$\bar{\phi}_x(\tau) = \frac{1}{T} \int_{-T/2}^{T/2} \phi_x(t + \tau, t) dt$$

For jointly WSS processes, the cross-spectrum is given by $S_{xy}(f) = \mathcal{F}[\phi_{xy}(\tau)]$.

The output of a LTI system with impulse response $h(t)$ to the WSS input $x(t)$ is the WSS process given by the convolution integral

$$y(t) = h(t) \otimes x(t) \triangleq \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

The two processes $x(t)$ and $y(t)$ are jointly WSS. The mean and autocorrelation of $y(t)$ and the cross-correlation between $x(t)$ and $y(t)$ are given by

$$\begin{aligned}\mu_y &= \mu_x \int_{-\infty}^{\infty} h(t) dt \\ \phi_y(\tau) &= \phi_h(\tau) \otimes \phi_x(\tau) \\ \phi_{xy}(\tau) &= h(-\tau)^* \otimes \phi_x(\tau)\end{aligned}$$

In the frequency domain we have

$$\begin{aligned}\mu_y &= \mu_x H(0) \\ S_y(f) &= |H(f)|^2 S_x(f) \\ S_{xy}(f) &= H(f)^* S_x(f)\end{aligned}$$

Since $\phi_{yx}(\tau) = \phi_{xy}(-\tau)^*$, we have $S_{yx}(f) = S_{xy}(f)^*$, that yields $S_{yx}(f) = H(f)S_x(f)$, since $S_x(f)$ is real.

1.2.3 Sampling bandlimited signals and processes

Let $x(t)$ be a bandlimited signal, i.e., such that $X(f)$ is zero for $f \notin [-W/2, W/2]$. We will now show that $x(t)$ is defined uniquely by its samples $x[i] = x(i/W) = x(t)\delta(t - i/W)$ taken at rate W , through the interpolation formula

$$x(t) = \sum_i x[i] \text{sinc}(W(t - i/W)) \quad (1.2)$$

where $\text{sinc}(t) \triangleq \frac{\sin(\pi t)}{\pi t}$. Sampling can be seen as the multiplication of $x(t)$ by a sequence of Dirac delta functions spaced by $1/W$ in the sense that

$$x_s(t) = x(t) \sum_i \delta(t - i/W) = \sum_i x[i] \delta(t - i/W). \quad (1.3)$$

In the frequency domain, $x_s(t)$ has a periodic transform since the Fourier transform of $\sum_i \delta(t - i/W)$ is given by $W \sum_i \delta(f - iW)$. It follows that

$$X_s(f) = \mathcal{F}(x_s(t)) = X(f) \otimes W \sum_i \delta(f - iW) = W \sum_i X(f - iW). \quad (1.4)$$

If W is larger than the maximum frequency for which $X(f)$ is non-zero, the replicas of $X(f)$ translated in integer multiples of W in the above periodic spectrum are non-overlapping. Therefore,

$$X(f) = \frac{1}{W} \Pi(f/W) W \sum_i X(f - iW) \quad (1.5)$$

where $\Pi(f)$ is the ideal brickwall low-pass filter with amplitude 1 for $f \in [-1/2, 1/2]$ and zero elsewhere. By applying inverse Fourier transform to the above equation, and by recalling that

$$\mathcal{F}^{-1}[\Pi(f)] = \text{sinc}(t)$$

we obtain the interpolation or reconstruction formula (1.2). The steps in (1.2,1.3,1.4,1.5) are shown pictorially in Figure 1.3.

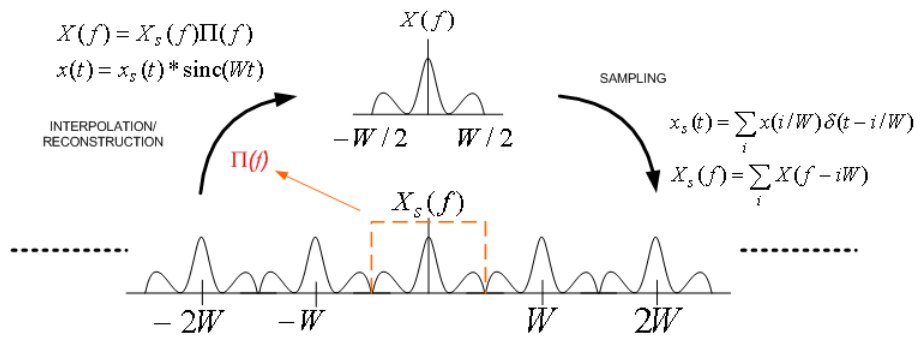


Figure 1.3: The Sampling and Reconstruction Processes

Discrete-time Fourier Transform (DFT)

The *discrete-time Fourier transform (DFT)* of the signal $x[i]$ is defined by

$$\mathcal{X}(\lambda) = \sum_i x[i] \exp(-j2\pi\lambda i) \quad (1.6)$$

for $\lambda \in [-1/2, 1/2]$. The relation between the continuous-time Fourier transform $X(f)$ of $x(t)$ and the discrete-time Fourier transform $\mathcal{X}(\lambda)$ of the sampled version $x[i]$ of $x(t)$ is called *spectral folding* and it is obtained as follows

$$\begin{aligned} \mathcal{X}(\lambda) &= \sum_i x[i] \exp(-j2\pi\lambda i) \\ &= \sum_i \int_{-\infty}^{\infty} X(f) \exp(j2\pi fi/W) df \exp(-j2\pi\lambda i) \\ &= \int_{-\infty}^{\infty} X(f) \sum_i \exp(j2\pi(f - W\lambda)i/W) df \\ &= \int_{-\infty}^{\infty} X(f) W \sum_i \delta(f - (\lambda + i)W) df \\ &= W \sum_i X((\lambda + i)W) \end{aligned} \quad (1.7)$$

where we have used the identity

$$\sum_i \exp(j2\pi ai/W) = W \sum_i \delta(a - iW)$$

A WSS process is said to be bandlimited in $[-W/2, W/2]$ if its PSD is zero outside this interval. In this case, the autocorrelation function of $x(t)$ can be written as

$$\phi_x(\tau) = \sum_i \phi_x[i] \text{sinc}(W(\tau - i/W))$$

and the interpolation formula (1.2) holds in the mean-square sense, i.e.,

$$E \left[\left| x(t) - \sum_i x[i] \text{sinc}(W(t - i/W)) \right|^2 \right] = 0$$

1.2.4 Representation of finite-energy signals by orthonormal expansions

The set of all signals $x(t)$ with finite energy, defined over the interval $-T/2, T/2$ (where T can be infinite), is a Hilbert space, i.e., a closed complete vector space with inner product

$$(x|y) \triangleq \int_{-T/2}^{T/2} x(t)^* y(t) dt$$

The norm of this space is given by $\|x\|^2 = (x|x)$, and the squared distance between two elements of the space is given by

$$\|x(t) - y(t)\|^2 = \int_{-T/2}^{T/2} |x(t) - y(t)|^2 dt = \|x\|^2 + \|y\|^2 - 2\text{Re}\{(x|y)\}$$

Two signals are said to be orthogonal if $(x|y) = 0$. A set of signals $\mathcal{B} = \{\xi_1(t), \dots, \xi_n(t)\}$ is said to be orthonormal if

$$(\xi_i|\xi_j) = \delta_{i,j}$$

In general Hilbert spaces, we have the following fundamental results

Cauchy-Schwarz inequality. For all x, y ,

$$|(x|y)| \leq \|x\| \|y\|$$

with equality if and only if $x(t) = \lambda y(t)$, for some $\lambda \in \mathbb{C}$. □

Projection theorem. Let S be a subspace of a Hilbert space, and x an arbitrary element of the Hilbert space. Then, there is a unique element $\tilde{x} \in S$ minimizing $\|x - y\|$ over all $y \in S$. Moreover, $\tilde{x}$ must satisfy the necessary and sufficient condition $(x - \tilde{x}|y) = 0$ for all $y \in S$. $\tilde{x}$ is called orthogonal projection of x on S . □

Gram-Schmidt orthonormalization. Let $x_1, \dots, x_n$ be elements of a Hilbert space, and let $S = \text{span}(x_1, \dots, x_n)$ be the subspace generated by $x_1, \dots, x_n$. Then, there exists an orthonormal set (basis) $\mathcal{B} = \{\xi_1, \dots, \xi_m\}$ with $m \leq n$, such that $S = \text{span}(\xi_1, \dots, \xi_m)$. □

The proof of the above statement is constructive and it is known as the *Gram-Schmidt algorithm* which allows one to find an orthonormal basis for the subspace $S = \text{span}(x_1, \dots, x_n)$ starting from the elements x_i . It is as follows: Let $j = 0$ and define $s_i = \sum_{\ell=1}^{j-1} (\xi_\ell|x_i) \xi_\ell$. Then, for $i = 1, \dots, n$

1. Let $z_i = x_i - s_i$.
2. If $z_i \neq 0$, then let $j \leftarrow j + 1$ and $\xi_j = z_i / \|z_i\|$.

In general, the basis obtained by the above algorithm depends on the order in which the elements x_i are considered. Different orderings yield different bases for the same vector space.

Minimum distance approximation. We want to approximate an arbitrary element x in a Hilbert space by a linear combination of elements $x_1, \dots, x_n$. In other words, we look for the element $\tilde{x} \in S = \text{span}(x_1, \dots, x_n)$ such that $\|x - \tilde{x}\|$ is minimum. In general, we can write $\tilde{x}$ as the linear combination

$$\tilde{x} = \sum_{i=1}^n \alpha_i x_i$$

By applying the projection theorem we get

$$\left(y \left| x - \sum_{i=1}^n \alpha_i x_i \right. \right) = 0$$

for all $y \in S$. The above equation holds in particular for all x_i , and since any $y \in S$ can be written as a linear combination of the x_i 's, by the linearity of the inner product we get that if it holds for all x_i , then it holds for all $y \in S$. Eventually, we obtain the system of equations

$$\begin{aligned} r_{1,1}\alpha_1 + r_{1,2}\alpha_2 + \dots + r_{1,n}\alpha_n &= d_1 \\ r_{2,1}\alpha_1 + r_{2,2}\alpha_2 + \dots + r_{2,n}\alpha_n &= d_2 \\ &\vdots \\ r_{n,1}\alpha_1 + r_{n,2}\alpha_2 + \dots + r_{n,n}\alpha_n &= d_n \end{aligned} \tag{1.8}$$

where we let $r_{i,j} = (x_i | x_j)$ and $d_i = (x_i | x)$. The above equations are called *normal equations* and can be written in compact form as $\mathbf{R}\boldsymbol{\alpha} = \mathbf{d}$, where $\mathbf{R} = [r_{i,j}]$ is the matrix of inner products of the elements x_i , and it is called *Gram matrix*, and $\mathbf{d}$ is the vector of the inner products between the x_i 's and the desired element x . The determinant of $\mathbf{R}$ is non-zero if and only if the elements x_i are linearly independent. If they are not, there exist multiple solutions. However, a possibly not unique solution always exists.

Infinite series of orthogonal elements. Let $\{\xi_1, \xi_2, \dots\}$ be an infinite set of orthonormal elements in a Hilbert space. The series $\sum_{i=1}^n \alpha_i \xi_i$ converges to an element x in the space if and only if $\sum_{i=1}^n |\alpha_i|^2 \leq \infty$. In that case, we have $\alpha_i = (\xi_i | x)$. $\square$

Bessel inequality. Let x be an arbitrary element of a Hilbert space, and let $\{\xi_1, \xi_2, \dots\}$ be an orthonormal sequence in the space, then

$$\sum_{i=1}^{\infty} |(\xi_i | x)|^2 \leq \|x\|^2$$

$\square$

From the projection theorem, we have that in general the series $\tilde{x} = \sum_{i=1}^{\infty} (\xi_i | x) \xi_i$ exists and it is the element of the closed subspace generated by the orthonormal sequence $\{\xi_1, \xi_2, \dots\}$ at minimum distance from x . Moreover, the *approximation error* $x - \tilde{x}$ is orthogonal to this subspace.

Complete orthonormal sequences. A sequence of orthonormal elements $\{\xi_1, \xi_2, \dots\}$ in a given Hilbert space is said to be complete if the closed subspace generated by it is the whole Hilbert space. Moreover, a sequence of orthonormal elements is complete if and only if the only element of the space orthogonal to each element of the sequence is 0. $\square$

Example: Fourier series. The sequence of orthonormal functions $\{\xi_i(t) = \frac{1}{\sqrt{T}}e^{j2\pi it/T} : i \in \mathbb{Z}\}$ is complete for the Hilbert space of finite-energy signals over the interval $[-T/2, T/2]$. The expansion of a given signal $x(t)$ in terms of the $\xi_i(t)$ is the well-known exponential Fourier series

$$x(t) = \sum_i \alpha_i e^{j2\pi it/T}$$

where the i -th Fourier coefficient is given by

$$\alpha_i = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-j2\pi it/T} dt$$

Geometrical representation of finite-dimensional signals. Let $x_1(t), \dots, x_n(t)$ be a set of signals (not necessarily linearly independent) and let $\mathcal{B} = (\xi_1(t), \dots, \xi_m(t))$ be an orthonormal basis for $S = \text{span}(x_1(t), \dots, x_n(t))$. Then, any signal $y(t) \in S$ is represented in a unique way as

$$y(t) = \sum_{i=1}^m y_i \xi_i(t)$$

where $y_i = (\xi_i | y)$. Therefore, there is a one-to-one correspondence between the signals in S and the complex m -dimensional vectors in $\mathbb{C}^m$. In other words, once we fix a basis of the signal set, we can represent signals as finite-dimensional vectors.

1.3 Baseband complex equivalent system

In general, a communication channel is characterized by a range of frequencies where its transfer function has non-negligible magnitude. The channel frequency band is specified by giving its center frequency (carrier) f_c and its one-sided bandwidth B . A signal $s(t)$ suited to travel along the channel must have his Fourier transform (spectrum) concentrated around f_c , over the bandwidth B . Therefore, it can be written as

$$\begin{aligned} s(t) &= a(t) \cos(2\pi f_c t + \theta(t)) \\ &= x_I(t) \cos(2\pi f_c t) - x_Q(t) \sin(2\pi f_c t) \\ &= \text{Re} \left\{ x(t) e^{j2\pi f_c t} \right\} \end{aligned} \tag{1.9}$$

where $a(t)$, $\theta(t)$, $x_I(t)$, $x_Q(t)$ and $x(t)$ are called amplitude, phase, in-phase component, quadrature component and complex envelope of the signal $s(t)$, respectively.

In the frequency domain, by letting $S(f) = \mathcal{F}[s(t)]$ and $X(f) = \mathcal{F}[x(t)]$, we have

$$\begin{aligned} S(f) &= \int_{-\infty}^{\infty} \operatorname{Re} \left\{ x(t) e^{j2\pi f_c t} \right\} e^{-j2\pi f t} dt \\ &= \frac{1}{2} \int_{-\infty}^{\infty} \left(x(t) e^{j2\pi f_c t} + x(t)^* e^{-j2\pi f_c t} \right) e^{-j2\pi f t} dt \\ &= \frac{1}{2} (X(f - f_c) + X(-f - f_c)^*) \end{aligned} \quad (1.10)$$

Since the frequency content of $s(t)$ is concentrated around $\pm f_c$, the frequency content of $x(t)$ is concentrated around frequency zero (baseband). That is why the complex envelope is also called *baseband complex equivalent* signal.

A linear system with real impulse response $h(t)$ has the symmetry property

$$H(-f)^* = H(f)$$

If the system is pass band with center frequency f_c , let

$$C(f - f_c) = H(f)u(f) \quad (1.11)$$

and

$$C(-f - f_c)^* = H(-f)^* u(-f) \quad (1.12)$$

where $u(f)$ is the unit step function. From the symmetry property of $H(f)$, we have that

$$H(f) = C(f - f_c) + C(-f - f_c)^*$$

that in the time-domain becomes

$$\begin{aligned} h(t) &= c(t) e^{j2\pi f_c t} + c(t)^* e^{-j2\pi f_c t} \\ &= 2 \operatorname{Re} \left\{ c(t) e^{j2\pi f_c t} \right\} \end{aligned} \quad (1.13)$$

The complex signal $c(t)$ is called *complex baseband equivalent* channel impulse response. Let $r(t) = h(t) \otimes s(t)$. In the frequency domain this becomes

$$\begin{aligned} R(f) &= \frac{1}{2} (X(f - f_c) + X(-f - f_c)^*) (C(f - f_c) + C(-f - f_c)^*) \\ &= \frac{1}{2} (X(f - f_c) C(f - f_c) + X(-f - f_c)^* C(-f - f_c)^*) \\ &= \frac{1}{2} (V(f - f_c) + V(-f - f_c)^*) \end{aligned} \quad (1.14)$$

where $V(f) = X(f)C(f)$ and where we let $r(t) = \operatorname{Re}\{v(t)e^{j2\pi f_c t}\}$. We conclude that the complex envelope $v(t)$ of the output $r(t)$ can be obtained by convolving the complex envelope $x(t)$ of the input by the channel baseband equivalent impulse response $c(t)$.

This fact allows us to work on complex baseband equivalent signals and systems, and neglect the carrier term. All what stated in terms of complex envelopes, can be easily translated in terms of the corresponding real modulated signals. A special case is represented by baseband signals and systems (for which $f_c = 0$). In this case, the real signal is equal to its in-phase component, and the quadrature component is zero. The complex envelope coincides with the signal itself and it is also real.

1.3.1 Passband WSS processes

Let $z(t)$ be a bandpass real WSS process. Then, we can write

$$\begin{aligned} z(t) &= \operatorname{Re} \left\{ n(t) e^{j2\pi f_c t} \right\} \\ &= n_I(t) \cos(2\pi f_c t) - n_Q(t) \sin(2\pi f_c t) \end{aligned} \quad (1.15)$$

It can be shown [3] that the quadrature components n_I and n_Q satisfy

$$\begin{aligned} \phi_{n_I}(\tau) &= \phi_{n_Q}(\tau) \\ \phi_{n_I, n_Q}(\tau) &= -\phi_{n_Q, n_I}(\tau) \end{aligned} \quad (1.16)$$

By using these symmetry properties, we obtain

$$\begin{aligned} \phi_z(\tau) &= \phi_{n_I}(\tau) \cos(2\pi f_c \tau) - \phi_{n_I, n_Q}(\tau) \sin(2\pi f_c \tau) \\ \phi_n(\tau) &= 2(\phi_{n_I}(\tau) + j\phi_{n_I, n_Q}(\tau)) \end{aligned} \quad (1.17)$$

that yield

$$\phi_z(\tau) = \frac{1}{2} \operatorname{Re} \left\{ \phi_n(\tau) e^{j2\pi f_c \tau} \right\}$$

In the frequency domain, by applying Fourier transform to the above relation we obtain

$$S_z(f) = \frac{1}{4} (S_n(f - f_c) + S_n(-f - f_c))$$

From the fact that for every two jointly WSS real processes, the cross-correlation function satisfies $\phi_{n_Q, n_I}(\tau) = \phi_{n_I, n_Q}(-\tau)$, and by using the above symmetry relation, we get that $\phi_{n_I, n_Q}(\tau)$ must be an odd function, i.e.,

$$\phi_{n_I, n_Q}(\tau) = -\phi_{n_I, n_Q}(-\tau)$$

Consequently, $\phi_{n_I, n_Q}(0) = 0$, i.e., the in-phase and quadrature components of a WSS bandpass process are uncorrelated for zero relative delay. This does not mean that $n_I(t_1)$ and $n_Q(t_2)$ are uncorrelated for all t_1 and t_2 .

1.3.2 White Gaussian noise

A particularly important noise model is the *white circularly-symmetric complex Gaussian* noise $n(t) = n_I(t) + jn_Q(t)$, where $n_I(t)$ and $n_Q(t)$ are jointly Gaussian zero-mean (real) processes, such that

$$\begin{aligned} E[n_I(t)n_I(t - \tau)] &= \frac{N_0}{2} \delta(\tau) \\ E[n_Q(t)n_Q(t - \tau)] &= \frac{N_0}{2} \delta(\tau) \\ E[n_I(t)n_Q(t - \tau)] &= 0 \end{aligned} \quad (1.18)$$

The autocorrelation function of the complex noise $n(t)$ is given by $E[n(t)n(t - \tau)^*] = N_0 \delta(\tau)$, and its PSD is constant and equal to N_0 over all frequencies.

White noise does not exist in nature, since it would have infinite power. However, it is a good model when the bandwidth of the noise introduced by the channel is much larger than the bandwidth of the signal. In any case, white noise is always “seen” through a filter, so that the filtered noise does have finite power, and it is physically meaningful.

In particular, we shall always deal with the following two cases:

Low-pass filtered and sampled noise. Let $\Psi(f)$ be a low-pass filter such that $\int_{-\infty}^{\infty} |\Psi(f)|^2 df < \infty$, and let $\psi(t)$ the corresponding impulse response. A discrete-time circularly-symmetric Gaussian process can be obtained by passing $n(t)$ through the filter $\Psi(f)$ and by sampling the output at a given rate W . We have

$$\nu[i] = \int_{-\infty}^{\infty} \psi(i/W - \tau) n(\tau) d\tau \quad (1.19)$$

The autocorrelation function of $\nu[i]$ is given by

$$\begin{aligned} \phi_{\nu}[i - j] &= E[\nu[i]\nu[j]^*] \\ &= E \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi(i/W - \tau) n(\tau) \psi(j/W - \theta)^* n(\theta)^* d\tau d\theta \right] \\ &= N_0 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi(i/W - \tau) \psi(j/W - \theta)^* \delta(\tau - \theta) d\tau d\theta \\ &= N_0 \int_{-\infty}^{\infty} \psi(\tau) \psi(\tau - (i - j)/W)^* d\tau \\ &= N_0 \phi_{\psi}((i - j)/W) \end{aligned} \quad (1.20)$$

The variance of the sampled noise is given by $\sigma_{\nu}^2 = N_0 \|\psi(t)\|^2$. In particular, if $\psi(t)$ has unit energy and its autocorrelation function satisfies $\phi_{\psi}(k/W) = 0$ for all $k \neq 0$, then $\nu[i]$ is a complex Gaussian discrete-time process such that $\nu_I[i] = \text{Re}\{\nu[i]\}$ and $\nu_Q[i] = \text{Im}\{\nu[i]\}$ are independent identically distributed Gaussian processes, with mean 0, cross-correlation 0 and autocorrelation function $N_0/2\delta_{k,0}$. We say that $\nu[i]$ is a *white* (i.e., uncorrelated) discrete-time complex circularly-symmetric Gaussian process with mean 0 and variance $E[|\nu[i]|^2] = N_0$. The term circularly-symmetric is due to the fact that the probability density function (pdf) of $\nu[i]$ and of $\nu[i]e^{j\theta}$ is the same, for any θ independent of $\nu[i]$. In other words, the pdf of $\nu[i]$ is invariant with respect to phase rotations.

Projected noise. Consider the projection of $n(t)$ over the set of signals $\{\xi_1(t), \dots, \xi_M(t)\}$ limited to $[-T/2, T/2]$. Let

$$\nu_i = \int_{-T/2}^{T/2} \xi_i(t)^* n(t) dt$$

The vector $\boldsymbol{\nu} = (\nu_1, \dots, \nu_M)^T$ is circularly-symmetric complex Gaussian with mean zero and covariance matrix $\mathbf{R}$ whose (i, j) -th element is given by

$$\begin{aligned} [\mathbf{R}]_{i,j} &= E[\nu_i \nu_j^*] \\ &= E \left[\int_{-T/2}^{T/2} \int_{-T/2}^{T/2} \xi_i(\tau)^* n(\tau) \xi_j(\theta) n(\theta)^* d\tau d\theta \right] \\ &= N_0 \int_{-T/2}^{T/2} \int_{-T/2}^{T/2} \xi_i(\tau)^* \xi_j(\theta) \delta(\tau - \theta) d\tau d\theta \\ &= N_0 \int_{-T/2}^{T/2} \xi_i(\tau)^* \xi_j(\tau) d\tau \end{aligned} \quad (1.21)$$

If $\{\xi_i(t)\}$ is an orthonormal set, then $\mathbf{R} = N_0 \mathbf{I}$ (i.e., $\boldsymbol{\nu}$ is white vector).

The proper complex Gaussian distribution. The joint probability density function (pdf) of the random variables ν_I and ν_Q , corresponding to the real and imaginary parts of a Gaussian noise sample ν , is given by

$$f_{\nu_I, \nu_Q}(x, y) = \frac{1}{\sqrt{\pi N_0}} \exp\left(-\frac{1}{N_0} x^2\right) \frac{1}{\sqrt{\pi N_0}} \exp\left(-\frac{1}{N_0} y^2\right) \quad (1.22)$$

We can define the complex variable $z = x + jy$, and use the short-hand notation

$$f_\nu(z) = \frac{1}{\pi N_0} \exp\left(-\frac{1}{N_0} |z|^2\right) \quad (1.23)$$

The probability distribution corresponding to the above pdf is called the one-dimensional *proper complex Gaussian distribution*. As already said above, we notice that the pdf of ν is invariant to rotations, i.e., $\nu' = e^{j\phi}\nu$ has the same distribution of ν for any deterministic phase ϕ . For this reason, the sampled (or projected) baseband-equivalent narrowband white Gaussian noise is said to be circularly-symmetric.

The proper complex Gaussian distribution with mean μ and variance σ^2 is indicated by $\mathcal{N}_{\mathbb{C}}(\mu, \sigma^2)$. Then, we use the notation $\nu \sim \mathcal{N}_{\mathbb{C}}(0, N_0)$ to indicate that ν is proper complex Gaussian with mean zero and variance N_0 .

More generally, an n -dimensional complex random vector $\mathbf{x}$ with joint pdf

$$f_{\mathbf{x}}(\mathbf{z}) = \frac{1}{\pi^n \det(\mathbf{R})} \exp\left(-(\mathbf{z} - \boldsymbol{\mu})^H \mathbf{R}^{-1} (\mathbf{z} - \boldsymbol{\mu})\right) \quad (1.24)$$

is said to be proper Gaussian, with mean $\boldsymbol{\mu} = E[\mathbf{x}]$ and covariance matrix $\mathbf{R} = E[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^H]$, and it is indicated as $\mathbf{x} \sim \mathcal{N}_{\mathbb{C}}(\boldsymbol{\mu}, \mathbf{R})$. Let $\mathbf{x} = \mathbf{a} + j\mathbf{b}$ with $\mathbf{a}$ and $\mathbf{b}$. Then, the joint pdf of $(\mathbf{a}, \mathbf{b})$ is Gaussian with mean $(\text{Re}\{\boldsymbol{\mu}\}, \text{Im}\{\boldsymbol{\mu}\})$ and covariance matrix

$$\tilde{\mathbf{R}} = \frac{1}{2} \begin{bmatrix} \text{Re}\{\mathbf{R}\} & -\text{Im}\{\mathbf{R}\} \\ \text{Im}\{\mathbf{R}\} & \text{Re}\{\mathbf{R}\} \end{bmatrix}$$

Notice that since $\mathbf{R}$ is Hermitian symmetric, then $\text{Re}\{\mathbf{R}\}$ is symmetric and $\text{Im}\{\mathbf{R}\}$ is antisymmetric, therefore $\tilde{\mathbf{R}}$ above is symmetric.

In general, any random vector $\mathbf{x}$ (also non-Gaussian) with covariance matrix satisfying the above structure is said to be *proper*. A proper random vector has zero pseudo-covariance matrix, i.e.,

$$E[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] = \frac{1}{2} (\text{Re}\{\mathbf{R}\} - \text{Re}\{\mathbf{R}\}) - \frac{j}{2} (\text{Im}\{\mathbf{R}\} + \text{Im}\{\mathbf{R}\}^T) = \mathbf{0}$$

1.4 Bayesian hypothesis testing

In this section we study the following problem:

Bayesian hypothesis testing. Consider two sets $\mathcal{X} = \{1, \dots, M\}$ and $\mathcal{Y}$, and a joint probability assignment defined on $\mathcal{X} \times \mathcal{Y}$. Let $(X, Y) \in \mathcal{X} \times \mathcal{Y}$ be a pair of joint random variables, and consider the M hypotheses:

$$\begin{aligned} H_1 &: Y \sim P_1(y) \\ H_2 &: Y \sim P_2(y) \\ &\vdots \\ H_M &: Y \sim P_M(y) \end{aligned} \quad (1.25)$$

where if $\mathcal{Y}$ is discrete, $P_x(y) \triangleq \Pr(Y = y|X = x)$, and if $\mathcal{Y}$ is continuous, $P_x(y) = \Pr(Y \leq y|X = x)$ with density $p_x(y)$.

By observing the event $\{Y = y\}$ we want to determine the best hypothesis $\hat{x}$ in order to minimize the *Bayes risk*

$$R = E[r(X, Y)]$$

where $r(x, y)$ is a given cost function of (x, y) . $\diamond$

For simplicity, in the following we use the notation for the case of continuous $\mathcal{Y}$. The operation of “guessing” the hypothesis by observing $\{Y = y\}$ is expressed mathematically by a function

$$g : \mathcal{Y} \rightarrow \{\mathcal{X}, e\}$$

defined on the whole $\mathcal{Y}$ called *decision rule*. The range of $g(y)$ is the set of hypotheses plus an extra symbol e , corresponding to the “I don’t know” answer (we assume that the cost function $r(x, y)$ is defined also for $x = e$). Since $\{\mathcal{X}, e\}$ is a discrete set of $M + 1$ elements, any decision rule $g(y)$ defines the *decision regions*

$$\mathcal{D}_i = \{y \in \mathcal{Y} : g(y) = i\} \quad \text{for } i = 1, \dots, M, e$$

Vice versa, any *partition* of $\mathcal{Y}$ into at most $M + 1$ regions $\mathcal{D}_i$ defines a decision rule

$$g(y) = i \quad \text{if } y \in \mathcal{D}_i$$

Fig. 1.4 illustrates the concept of the decision regions.

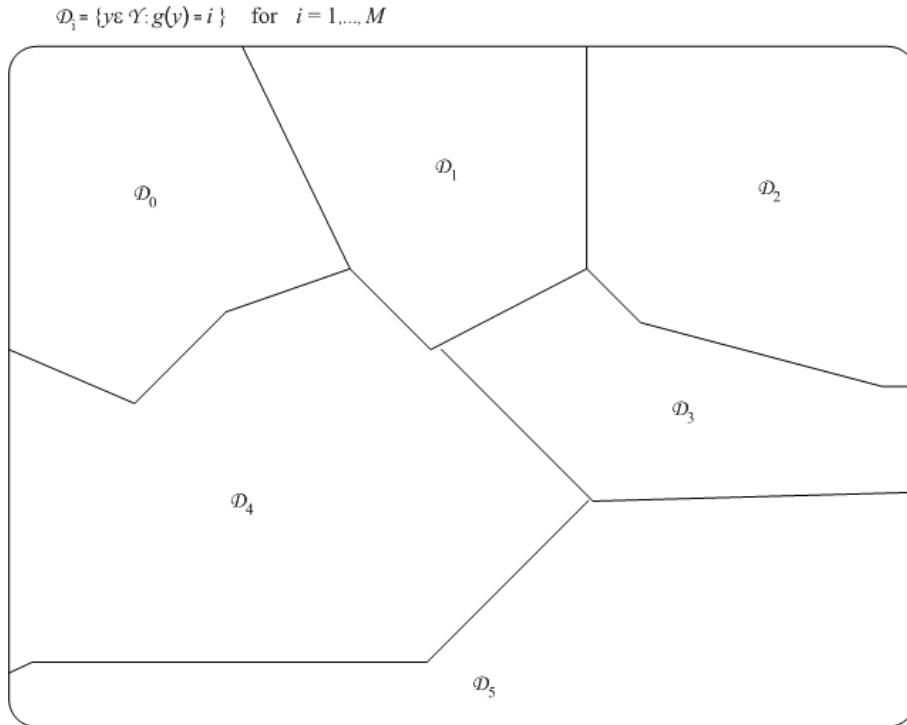


Figure 1.4: Partitioning of the observation set $\mathcal{Y}$ by the decision regions.

The most important Bayes risk in classical communications problems is the average error probability, given by

$$P(e) = \Pr(g(Y) \neq X) \quad (1.26)$$

which corresponds to the choice of the cost function

$$r(x, y) = 1\{g(y) \neq x\}$$

for $x = 1, \dots, M$. It follows immediately that any decision rule minimizing $P(e)$ has necessarily $\mathcal{D}_e = \emptyset$. In other words, the answer “I don’t know” never helps, if the goal is to minimize the average error probability, since its associated cost is always 1 error.

In order to determine the optimal decision rule, we can write

$$\begin{aligned} P(e) &= \Pr(g(Y) \neq X) \\ &= 1 - \Pr(g(Y) = X) \\ &= 1 - \sum_{x=1}^M \Pr(X = x) \Pr(g(Y) = x | X = x) \\ &= 1 - \sum_{x=1}^M \int_{\mathcal{D}_x} p_x(y) \pi(x) dy \end{aligned} \quad (1.27)$$

where we let $\pi(x) = \Pr(X = x)$ be the *a priori* probability of the hypothesis x .

The optimal decision rule $g(y)$ has decision regions $\mathcal{D}_x$ maximizing the sum of integrals in the last line of (1.27). Since $\{\mathcal{D}_x\}$ is a partition of $\mathcal{Y}$, the same differential element $p_x(y)\pi(x)dy$ cannot be assigned to more than one region. Therefore, the optimal choice for the regions is to assign to $\mathcal{D}_x$ all points $y \in \mathcal{Y}$ such that $p_x(y)\pi(x)$ is not smaller than any $p_i(y)\pi(i)$, for all $i = 1, \dots, M$. Eventually, we obtain the optimal decision regions given by

$$\mathcal{D}_x = \left\{ y \in \mathcal{Y} : p_x(y)\pi(x) = \max_i \{p_i(y)\pi(i)\} \right\} \quad (1.28)$$

The boundary between two regions $\mathcal{D}_x$ and $\mathcal{D}_{x'}$ is given by the set of points y such that

$$p_x(y)\pi(x) = p_{x'}(y)\pi(x') = \max_i \{p_i(y)\pi(i)\}$$

and can be assigned to either $\mathcal{D}_x$ or $\mathcal{D}_{x'}$ without affecting the value of $P(e)$. Also, randomized decision rules choosing at random between x and x' if y belongs to the boundary yield the same average error probability.

The optimal decision rule defined by (1.28) can be written in a more concise form as

$$g(y) = \arg \max_{x \in \mathcal{X}} \{p_x(y)\pi(x)\} \quad (1.29)$$

This is sometimes referred to as *Maximum A Posteriori probability* (MAP) rule, since maximizing $p_x(y)\pi(x)$ with respect to x is equivalent to maximizing the a posteriori probability $\Pr(X = x | Y = y)$ of X given the the observation Y , in fact, from the Bayes rule we can write

$$\Pr(X = x | Y = y) = \frac{p_x(y)\pi(x)}{p(y)} = \frac{p_x(y)\pi(x)}{\sum_{i=1}^M p_i(y)\pi(i)}$$

(since the denominator in the RHS of the above equation is independent of x , maximization of $\Pr(X = x|Y = y)$ and of $p_x(y)\pi(x)$ yields the same decision regions).

A particularly important special case occurs when $\pi(x) = 1/M$ for all $x \in \mathcal{X}$ (uniform a priori probability). In this case, the term $\pi(x)$ can be neglected from maximization in (1.29), and the optimal rule consists of maximizing the conditional probability density $p_x(y)$ over $x \in \mathcal{X}$. In statistical estimation, the conditional density of the observation given the “parameter” that we want to estimate is called *likelihood function*, and the decision rule

$$g(y) = \arg \max_{x \in \mathcal{X}} \{p_x(y)\} \quad (1.30)$$

is called *Maximum-Likelihood* (ML) rule. In many practical cases, if the a priori probabilities are not known, they are artificially assumed to be uniform and the ML instead of the MAP rule is used.

Example: binary symmetric channel. Let $\mathcal{X} = \mathcal{Y} = \{0, 1\}$. Fig. 1.5 represents the diagram of a binary symmetric channel (BSC) with cross-over probability $p < 1/2$. Assuming uniform prior probabilities, we have

$$\begin{aligned} P_0(y) &= \begin{cases} 1-p & y=0 \\ p & y=1 \end{cases} \\ P_1(y) &= \begin{cases} p & y=0 \\ 1-p & y=1 \end{cases} \end{aligned}$$

The MAP (or, equivalently, the ML) rule is given by

$$g(y) = y$$

Obviously, if $p > 1/2$, the rule would be $g(y) = 1 \oplus y$. □

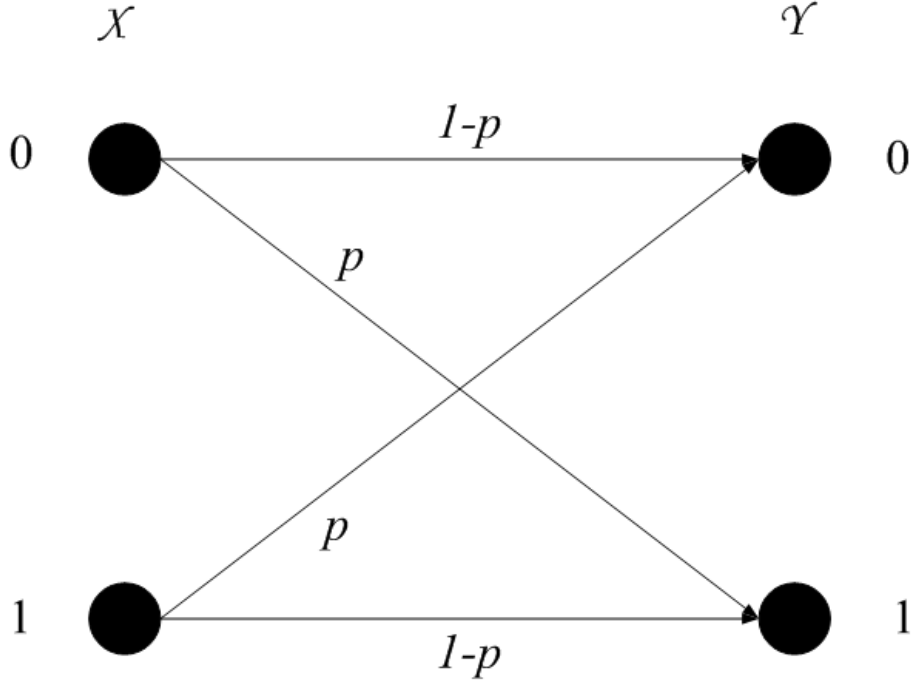
Example: location testing with Gaussian error. Let $\mathcal{X} = \{0, 1\}$ and $\mathcal{Y} = \mathbb{R}$. Assume that $Y \sim \mathcal{N}(\mu_x, \sigma^2)$, where $\mu_0 > \mu_1$ are given real numbers and where X has uniform prior probabilities. The MAP (or, equivalently, the ML) rule is defined by the decision region

$$\mathcal{D}_0 = \{y \in \mathbb{R} : p_0(y) \geq p_1(y)\}$$

The inequality defining $\mathcal{D}_0$ can be written as

$$\begin{aligned} p_0(y) &\geq p_1(y) \\ \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{|y-\mu_0|^2}{2\sigma^2}} &\geq \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{|y-\mu_1|^2}{2\sigma^2}} \\ |y-\mu_0|^2 &\leq |y-\mu_1|^2 \\ -2\mu_0 y + \mu_0^2 &\leq -2\mu_1 y + \mu_1^2 \\ 2(\mu_0 - \mu_1)y &\geq \mu_0^2 - \mu_1^2 \\ y &\geq (\mu_0 + \mu_1)/2 \end{aligned}$$

Therefore, the optimal decision rule consists of comparing the observation y with the threshold $(\mu_0 + \mu_1)/2$, which is equal to the middle point between μ_1 and μ_0 . □

Figure 1.5: The diagram of a BSC with cross-over probability p .

1.5 Upper and lower bounds to ML error probability

In this section we consider uniform prior probabilities and we find upper and lower bounds to $P(e)$ resulting from the (optimal) ML decision rule.

For any given pair of symbols $x \neq x'$, we define the pairwise error event

$$\{x \rightarrow x'\} \triangleq \{y \in \mathcal{Y} : p_x(y) \leq p_{x'}(y)\}$$

and the pairwise error probability (PEP) as

$$P(x \rightarrow x') \triangleq \Pr(\{x \rightarrow x'\} | X = x)$$

In words, $P(x \rightarrow x')$ is the probability that the ML decision rule chooses for x' when the true hypothesis was x , and as if x and x' were the two only possible hypotheses.

The event $\{g(y) \neq x\}$ (the complement set of $\mathcal{D}_x$) can be written as the union of pairwise error events

$$\{g(y) \neq x\} = \bigcup_{x' \neq x} \{x \rightarrow x'\}$$

Then, by using the fact that the probability of the union of events is not larger than the sum of

the probabilities of all individual events, we can write the upper bound

$$\begin{aligned}
 \Pr(g(Y) \neq x | X = x) &= \Pr\left(\bigcup_{x' \neq x} \{x \rightarrow x'\} | X = x\right) \\
 &\leq \sum_{x' \neq x} \Pr(\{x \rightarrow x'\} | X = x) \\
 &= \sum_{x' \neq x} P(x \rightarrow x')
 \end{aligned} \tag{1.31}$$

By averaging over all possible x (with respect to the uniform prior probability), we obtain the upper *union bound* on $P(e)$,

$$P(e) \leq \frac{1}{M} \sum_{x=1}^M \sum_{x' \neq x} P(x \rightarrow x') \tag{1.32}$$

We can obtain a looser but sometimes easier to calculate upper bound by upperbounding further all PEP terms in (1.32). Suppose that we find a function $f(y)$ such that

$$f(y) \geq 1\{p_x(y) \leq p_{x'}(y)\}$$

Then, we have

$$\begin{aligned}
 P(x \rightarrow x') &= \Pr(\{x \rightarrow x'\} | X = x) \\
 &= E[1\{p_x(Y) \leq p_{x'}(Y)\} | X = x] \\
 &\leq E[f(Y) | X = x]
 \end{aligned} \tag{1.33}$$

A particularly interesting choice for the function $f(y)$ is the following:

$$f(y) = \exp(-\lambda L_{x,x'}(y)) \tag{1.34}$$

where we define the log-likelihood ratio (LLR) between the hypotheses x and x' as

$$L_{x,x'}(y) \triangleq \log \frac{p_x(y)}{p_{x'}(y)}$$

and where $\lambda \geq 0$ is a parameter that can be further optimized in order to obtain a bound as tight as possible. The resulting bound is referred to as *Chernoff bound*, and it is given by

$$P(x \rightarrow x') \leq \min_{\lambda \geq 0} E \left[e^{-\lambda L_{x,x'}(Y)} \middle| X = x \right] \tag{1.35}$$

Interestingly, we can define a random variable $L \triangleq L_{x,x'}(Y)$ with probability distribution induced by $Y \sim p_x(y)$, and the expectation in (1.35) is the characteristic function

$$\Phi_L(s) \triangleq E[e^{-sL}]$$

of L evaluated over the real axis $s = \lambda \in \mathbb{R}$. In many important cases, this characteristic function is easier to calculate than the exact PEP.

From the trivial inclusion relation

$$\{g(y) \neq x\} \supseteq \{x \rightarrow x'\}$$

we obtain

$$\begin{aligned} \Pr(g(Y) \neq x | X = x) &\geq \max_{x' \neq x} \Pr(\{x \rightarrow x'\} | X = x) \\ &= \max_{x' \neq x} P(x \rightarrow x') \end{aligned} \quad (1.36)$$

By averaging over all possible x (with respect to the uniform prior probability), we obtain the lower bound on $P(e)$,

$$P(e) \geq \frac{1}{M} \sum_{x=1}^M \max_{x' \neq x} P(x \rightarrow x') \quad (1.37)$$

Unfortunately, we cannot use in the above lower bound the Chernoff upper bound on the PEP, since an upper bound of a lower bound is not a bound any longer.

In several cases of interest, the PEP is very easy to calculate while $P(e)$ is difficult if not impossible. Therefore, the upper bound (1.32) and the lower bound (1.37) are useful tools to estimate the error probability.

1.6 Sufficient statistics

Suppose that there exist a random variable Z such that the distribution of the observation Y conditioned on Z does not depend any longer on X , i.e., that

$$P(x, y, z) = P(y|x, z)P(x, z) = P(y|z)P(x, z)$$

where $P(x, y, z)$ is the joint probability assignment of X, Y, Z . Then, all the information that we can obtain on X by observing Y , is *captured* by Z . We say that Z is a *sufficient statistic* for X . A particular important case is when $Z = T(Y)$, i.e., when Z is a function of the observation. In other words, we can transform the observation Y by a function $T(y)$ without losing any useful information for the determination of X , provided that $Z = T(Y)$ is sufficient for X . We have the following result [2]:

Factorization theorem. Let $p_x(y)$ be the conditional density of Y given $X = x$. A statistic $T(y)$ is sufficient for X if and only if there exist functions $f_x(y)$ and $h(y)$ such that

$$p_x(y) = f_x(T(y)) h(y)$$

□

Example: sufficiency of the LLR in binary hypothesis testing. Consider the binary hypothesis testing problem with conditional probability densities $p_1(y)$ and $p_2(y)$. We define the LLR

$$L_{2,1}(y) = \log \frac{p_2(y)}{p_1(y)}$$

From the factorization theorem, we see immediately that $T(Y) = L_{2,1}(Y)$ is sufficient for X , in fact we are able to write

$$p_x(y) = \begin{cases} p_1(y) & x = 1 \\ e^{L_{2,1}(y)} p_1(y) & x = 2 \end{cases}$$

which is in the form $p_x(y) = f_x(T(y))h(y)$ with the choice $h(y) = p_1(y)$ and

$$f_x(t) = \begin{cases} 1 & x = 1 \\ e^t & x = 2 \end{cases}$$

1.7 Problems

Problem 1. Define $p_T(t)$ as

$$p_T(t) = \begin{cases} \frac{1}{\sqrt{T}} & 0 \leq t \leq T \\ 0 & \text{elsewhere} \end{cases}$$

and assume $f_c T \gg 1$.

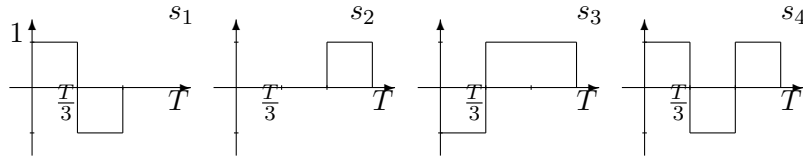
- 1) Find an orthonormal basis for the real signal set $\mathcal{S} = \{s_m(t) : m = 0, \dots, 7\}$ defined by

$$s_m(t) = \begin{cases} \operatorname{Re}\{p_T(t) \exp(j2\pi(f_c t + m/4))\} & m = 0, 1, 2, 3 \\ \sqrt{2} \operatorname{Re}\{p_T(t) \exp(j2\pi(f_c t + (m-4)/4 + 1/8))\} & m = 4, 5, 6, 7 \end{cases}$$

- 2) Represent the signal constellation corresponding to $\mathcal{S}$ in the real Euclidean space with respect to the basis found.

- 3) Find the average energy per signal, assuming that all signals are equiprobable.

Problem 2. Find an orthonormal basis for the four real signals given below:



and represent them as points in the Euclidean space generated by the basis found.

Then, compute the average energy per signal (assume that the signals are used with equal probability) and the minimum squared Euclidean distance between the signals.

Problem 3. Give a geometrical representation of the following set of signals

$$s_m(t) = \begin{cases} \operatorname{Re}\{A p_T(t) e^{j(\frac{\pi}{2}m + \frac{\pi}{4})} e^{j2\pi f_0 t}\} & \text{for } m = 0, 1, 2, 3 \\ \operatorname{Re}\{\frac{1+\sqrt{3}}{\sqrt{2}} A p_T(t) e^{j\frac{\pi}{2}(m-4)} e^{j2\pi f_0 t}\} & \text{for } m = 4, 5, 6, 7 \end{cases}$$

where A is real positive and $p_T(t)$ is a unit energy rectangular pulse of support $[0, T]$. Assume $f_0 T \gg 1$.

Problem 4. Represent the following signals $s_1(t)$, $s_2(t)$, $s_3(t)$ as points in a Euclidean geometrical space, by choosing an appropriate orthonormal basis.

$$\begin{aligned} s_1(t) &= \begin{cases} e^{-t} & t \in [0, 1] \\ 0 & \text{elsewhere} \end{cases} \\ s_2(t) &= \begin{cases} +1 & t \in [0, 1/2] \\ -1 & t \in (1/2, 1] \\ 0 & \text{elsewhere} \end{cases} \\ s_3(t) &= \begin{cases} -1 & t \in [0, 1/4] \\ +1 & t \in (1/4, 1] \\ 0 & \text{elsewhere} \end{cases} \end{aligned}$$

Problem 5. Consider an additive-noise channel $y = x + n$ where x takes on the values $\pm A$ with equal probability and where n is a Laplace random variable with probability density function

$$p_n(z) = \frac{1}{\sqrt{2}\sigma} e^{-|z|\sqrt{2}/\sigma}$$

Determine the decision regions of the MAP detector. Compare the decision regions found with those of the MAP detector for $n \sim \mathcal{N}(0, \sigma^2)$. Compute the error probability in the two cases of Laplace and Gaussian noise and compare the resulting error probabilities for the same SNR (note: the SNR here is defined as the average energy per symbol divided by the noise variance). What is the worst-case noise?

Problem 6. Consider a real binary input Laplacian noise channel $y = x + z$ with input alphabet $\{-A, +A\}$ and noise pdf

$$p(z) = \frac{b}{2} \exp(-b|z|)$$

The prior input probability distribution is $\Pr(x = A) = 0.8$, $\Pr(x = -A) = 0.2$. Find the MAP decision regions and compute the minimum average decision error probability.

Problem 7. Consider the signal $y = \sqrt{E}x + z + \nu$ where $x \in \{\pm 1\}$ represents the transmitted signal, $z \in \{\pm 1\}$ an interfering signal and ν is a real noise variable (independent of x and z) with distribution given by

$$p_\nu(u) = \begin{cases} 0 & \text{for } |u| \geq 1 \\ u + 1 & \text{for } -1 < u < 0 \\ -u + 1 & \text{for } 0 \leq u < 1 \end{cases}$$

x and z are independent and takes on the values $+1$ and -1 with equal probability.

Find the optimal MAP decision regions and the corresponding minimum symbol error probability for the detection of x with observation y .

Problem 8. Let $p(y|x)$ and $p(y|x')$ be the conditional pdfs of the observation y given the two equiprobable hypotheses x and x' . Prove the upper bound to the PEP

$$P(x \rightarrow x') \leq E \left[\sqrt{\frac{p(y|x')}{p(y|x)}} \middle| x \right] = \int \sqrt{p(y|x)p(y|x')} dy$$

Then, show that in the case where $y = x + \nu$, with $\nu \sim \mathcal{N}_{\mathbb{C}}(0, N_0)$ and x, x' are points in the complex plane, the above bound yields the same result of the Chernoff bound optimized with respect to the parameter λ for the hypothesis testing problem of detecting x by observing y .

Problem 9. A discrete communication channel with input x and output y has input alphabet $\{0, 1\}$, output alphabet $\{a, b, c, d\}$ and transition probability $P(y|x)$ given by

$$\begin{aligned} P(a|0) &= 0.3, & P(b|0) &= 0.1, & P(c|0) &= 0.5, & P(d|0) &= 0.1, \\ P(a|1) &= 0.1, & P(b|1) &= 0.5, & P(c|1) &= 0.1, & P(d|1) &= 0.3, \end{aligned}$$

The a priori probability of the input is $Q(x = 0) = 0.6$, $Q(x = 1) = 0.4$. Compute the maximum a posteriori (MAP) detection rule and the resulting average error probability.

Chapter 2

Signal detection in AWGN

In this chapter we study the elementary communication system described by the block diagram of Fig. ???. The transmitter (modulator) has a set $\mathcal{S}$ of M waveforms

$$\mathcal{S} = \{s_m(t) : m = 1, \dots, M\}$$

of finite energy and duration T . It generates an information message m at random with a given a priori probability distribution $\pi(m)$, and transmits the corresponding waveform $s_m(t)$. The transmission channel is distortionless and introduces additive white Gaussian noise (AWGN) $\nu(t)$ with power spectral density N_0 and a phase rotation θ . The phase rotation is assumed to be independent of the transmitted signal and of the noise, random but constant over the duration T of the waveform. The phase rotation represents the phase offset between the local oscillator of the transmitter and of the receiver (including the effect of the propagation through the channel). The received signal over the observation interval $[0, T]$ is

$$r(t) = s_m(t)e^{j\theta} + \nu(t) \quad (2.1)$$

The receiver must estimate with minimal probability of error the value of the transmitted message by observing the channel output $r(t)$, $t \in [0, T]$.

We shall distinguish the cases of *coherent* and *non-coherent* detection. In the first case, the receiver has perfect knowledge of the phase rotation θ . Therefore, it can apply an inverse rotation by multiplying $r(t)$ by $e^{-j\theta}$. Since the noise is rotationally invariant, the coherent detection channel model reduces to

$$r(t) = s_m(t) + \nu(t) \quad (2.2)$$

In practice, coherent detection can be obtained by estimating the signal phase θ . This can be achieved by some phase synchronization technique.

In the case where phase estimation is not possible or, in order to keep the receiver as simple as possible, it is not implemented, we are in the presence of non-coherent detection. In this case, we shall assume that the phase θ is completely unknown, and model it as an independent random variable uniformly distributed on $[-\pi, \pi)$.

Figure 2.1: Waveform transmission over the AWGN channel.

2.1 Coherent detection of binary signal sets

Before considering the general case of M arbitrary waveforms, we consider the simpler example of $M = 2$, where

$$s_m(t) = a_m \psi(t) \quad m = 1, 2$$

where $a_1 = +1$ and $a_2 = -1$ are called *modulation symbols* and $\psi(t)$ is an elementary waveform with energy $\mathcal{E} = \|\psi(t)\|^2$, used to convey information. This type of modulation is called binary (since there are two waveforms), and antipodal (since the two waveforms are mutually opposite). Moreover, since all signals in $\mathcal{S}$ are proportional to the same elementary waveform (they are linearly dependent), this type of modulation belongs to the class of so called *linear modulations*. The two messages are transmitted with probabilities $\pi(1)$ and $\pi(2)$, respectively.

In order to find the optimal detector yielding minimum error probability, we need to apply the MAP decision rule derived for the general hypothesis testing problem. However, it is not intuitive how to apply this to a continuous-time observation (see [2] for a complete and rigorous treatment). Therefore, we first find a discrete sufficient statistic for m and then apply the MAP detector to this statistics.

Let $\xi_1(t) = \psi(t)/\sqrt{\mathcal{E}}$ and let $\mathcal{B} = \{\xi_i(t) : i = 1, 2, \dots\}$ be a complete orthonormal basis for the finite-energy signals defined on $[0, T]$. The received signal can be written as

$$r(t) = \sum_{i=1}^N r_i \xi_i(t)$$

where

$$r_i = (\xi_i(t)|r(t)) = \int_0^T \xi_i(t)^* r(t) dt \quad (2.3)$$

Because of the construction of the basis $\mathcal{B}$, and from the result on projected white noise, we can write the coefficients r_i as

$$r_i = \begin{cases} a_m \sqrt{\mathcal{E}} + \nu_1 & i = 1 \\ \nu_i & i > 1 \end{cases}$$

where the projected noise samples ν_i are i.i.d. circularly symmetric complex Gaussian with variance N_0 . Obviously, the sequence of samples $\mathbf{r} = (r_1, r_2, \dots)$ is conditionally independent of the message m given its first sample r_1 . Therefore, we define $y = T(\mathbf{r}) = r_1$, which is sufficient for the detection of m .

The resulting discrete channel model is given by

$$y = a_m \sqrt{\mathcal{E}} + \nu \quad (2.4)$$

where $\nu \sim \mathcal{N}_c(0, N_0)$.

The conditional pdf of the observation y given m is

$$p(y|m) = \frac{1}{\pi N_0} \exp\left(-\frac{1}{N_0} |y - a_m \sqrt{\mathcal{E}}|^2\right) \quad (2.5)$$

Since we have to do with a binary hypothesis testing problem, the optimal decision rule is determined simply by the decision region for one of the hypotheses (the other is given by the complement region). Without loss of generality, we focus on the region $\mathcal{D}_1$. This is given by all $y \in \mathbb{C}$ satisfying the inequality

$$\pi(1) \frac{1}{\pi N_0} \exp\left(-\frac{1}{N_0} |y - \sqrt{\mathcal{E}}|^2\right) \geq \pi(2) \frac{1}{\pi N_0} \exp\left(-\frac{1}{N_0} |y + \sqrt{\mathcal{E}}|^2\right)$$

For simplicity, we let $\pi(1) = q$ and $\pi(2) = 1 - q$, and we take the logarithm of both sides. We obtain

$$\begin{aligned} \log q - \frac{1}{N_0} (|y|^2 - 2\sqrt{\mathcal{E}} \operatorname{Re}\{y\} + \mathcal{E}) &\geq \log(1 - q) - \frac{1}{N_0} (|y|^2 + 2\sqrt{\mathcal{E}} \operatorname{Re}\{y\} + \mathcal{E}) \\ \operatorname{Re}\{y\} &\geq \frac{N_0}{4\sqrt{\mathcal{E}}} \log \frac{1 - q}{q} \end{aligned}$$

Then, the optimal decision rule corresponds to comparing $\operatorname{Re}\{y\}$ with the threshold $\tau = \frac{N_0}{4\sqrt{\mathcal{E}}} \log \frac{1 - q}{q}$. If the prior probability is uniform (i.e., $q = 1 - q = 1/2$), the threshold is $\tau = 0$.

The resulting minimum probability of error is given by

$$P(e) = qP(e|m = 1) + (1 - q)P(e|m = 2)$$

where the conditional probability of error is given by

$$P(e|m = i) = \begin{cases} \Pr(\sqrt{\mathcal{E}} + \operatorname{Re}\{\nu\} < \tau) & i = 1 \\ \Pr(-\sqrt{\mathcal{E}} + \operatorname{Re}\{\nu\} > \tau) & i = 2 \end{cases}$$

First, we examine the conditional probability of error for message 1. We have

$$\begin{aligned}
 P(e|m=1) &= \Pr(\sqrt{\mathcal{E}} + \text{Re}\{\nu\} < \tau) \\
 &= \int_{-\infty}^{\tau - \sqrt{\mathcal{E}}} \frac{1}{\sqrt{\pi N_0}} e^{-z^2/N_0} dz \\
 &= \int_{\sqrt{\mathcal{E}} - \tau}^{+\infty} \frac{1}{\sqrt{\pi N_0}} e^{-z^2/N_0} dz \\
 &= Q\left(\frac{\sqrt{\mathcal{E}} - \tau}{\sqrt{N_0/2}}\right)
 \end{aligned} \tag{2.6}$$

where we define the Gaussian tail function

$$Q(z) \triangleq \int_z^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$$

Analogously, the conditional probability of error for message 2 is given by

$$\begin{aligned}
 P(e|m=2) &= \Pr(-\sqrt{\mathcal{E}} + \text{Re}\{\nu\} > \tau) \\
 &= \int_{\sqrt{\mathcal{E}} + \tau}^{+\infty} \frac{1}{\sqrt{\pi N_0}} e^{-z^2/N_0} dz \\
 &= Q\left(\frac{\sqrt{\mathcal{E}} + \tau}{\sqrt{N_0/2}}\right)
 \end{aligned} \tag{2.7}$$

The average error probability is given by

$$P(e) = qQ\left(\frac{\sqrt{\mathcal{E}} - \tau}{\sqrt{N_0/2}}\right) + (1-q)Q\left(\frac{\sqrt{\mathcal{E}} + \tau}{\sqrt{N_0/2}}\right)$$

which for uniform prior probability boils down to

$$P(e) = Q\left(\sqrt{\frac{2\mathcal{E}}{N_0}}\right)$$

The derivation of the above optimal MAP detector for binary antipodal signals shows that the detector can be divided into two distinct operations:

1. Projection of the received signal onto a basis of the signal space of the transmitter.
2. Decision rule based on the projected (sufficient) statistics.

We shall see later that this partitioning holds in general. The extraction of y from the received signal $r(t)$ is obtained simply by the projection (inner product) $y = (\xi_1(t)|r(t))$. Equivalently, this can be seen as a filtering and sampling operation, where $r(t)$ is passed through a filter with impulse response $h(t) = \xi_1(T-t)^*$, and the filter output is sampled at time T . In fact, we have

$$\begin{aligned}
 r(t) \otimes h(t)|_{t=T} &= \int h(t-\tau)r(\tau)d\tau \Big|_{t=T} \\
 &= \int \xi_1(T+\tau-t)^*r(\tau)d\tau \Big|_{t=T} \\
 &= \int \xi_1(\tau)^*r(\tau)d\tau \\
 &= (\xi_1(t)|r(t))
 \end{aligned}$$

The filter $h(t) = \xi_1(T - t)^*$ is said to be *matched* to the waveform $\xi_1(t)$. The properties of the matched filter will be seen later in this chapter. Figs. ?? and ?? represent the block diagrams of the optimal MAP detector for binary antipodal signals based on correlation and on matched filtering, respectively.

Figure 2.2: MAP detector for binary antipodal signals based on a signal correlator.

2.1.1 Matched filtering

We have seen before that a sufficient statistic for ML (or MAP) detection of m is obtained as the matched filter output. In the detection of signals in white noise, matched filtering plays a very important role [2]. In particular, we consider the signal model

$$r(t) = s(t) + \nu(t) \quad t \in [0, T]$$

where $\nu(t)$ is a non-necessarily Gaussian white noise with autocorrelation function $E[\nu(t_1)\nu(t_2)^*] = N_0\delta(t_1 - t_2)$, and we constrain the receiver to be formed by a linear filter $h(t)$ whose output is sampled at time T . The output sample is given by

$$y = \int_0^T h(T - \tau)r(\tau)d\tau$$

We define the signal-to-noise ratio (SNR) as the ratio between the squared magnitude of the signal component and the variance of the noise component at the filter output. Because of linearity, we have $y = x + z$, where

$$x = \int_0^T h(T - \tau)s(\tau)d\tau$$

Figure 2.3: MAP detector for binary antipodal signals based on a matched filter.

and

$$z = \int_0^T h(T - \tau) \nu(\tau) d\tau$$

Therefore, the SNR is given by

$$\begin{aligned} \text{SNR} &= \frac{|x|^2}{E[|z|^2]} \\ &= \frac{\left| \int_0^T h(T - \tau) s(\tau) d\tau \right|^2}{E \left[\left| \int_0^T h(T - \tau) \nu(\tau) d\tau \right|^2 \right]} \end{aligned}$$

Consider the denominator of the above expression. We can write

$$\begin{aligned} E \left[\left| \int_0^T h(T - \tau) \nu(\tau) d\tau \right|^2 \right] &= E \left[\int_0^T \int_0^T h(T - \tau) h(T - \theta)^* \nu(\tau) \nu(\theta)^* d\tau d\theta \right] \\ &= N_0 \int_0^T \int_0^T h(T - \tau) h(T - \theta)^* \delta(\tau - \theta) d\tau d\theta \\ &= N_0 \int_0^T |h(T - \tau)|^2 d\tau \end{aligned}$$

We have the following result:

SNR optimality of the matched filter. The maximum SNR over all possible filters $h(t)$ is $\text{SNR} = \frac{E_s}{N_0}$ where E_s is the energy of $s(t)$, and it is obtained by the matched filter $h(t) = s(T - t)^*$.

Proof. We write the output SNR as

$$\text{SNR} = \frac{\left| \int_0^T h(T-\tau)s(\tau)d\tau \right|^2}{N_0 \int_0^T |h(T-\tau)|^2 d\tau}$$

From the Cauchy-Schwartz inequality we have

$$\left| \int_0^T h(T-\tau)s(\tau)d\tau \right|^2 \leq \left(\int_0^T |h(T-\tau)|^2 d\tau \right) \left(\int_0^T |s(\tau)|^2 d\tau \right)$$

with equality if and only if $h(T-\tau) \propto s(\tau)^*$. Then, by letting $h(t) = s(T-t)^*$ we obtain the maximum SNR as

$$\text{SNR} = \frac{\left| \int_0^T |s(\tau)|^2 d\tau \right|^2}{N_0 \int_0^T |s(\tau)|^2 d\tau} = \frac{E_s}{N_0}$$

□

2.2 Coherent detection of M -ary signal sets

We are ready to consider the general problem where the modulator is equipped with an M -ary signal set $\mathcal{S} = \{s_1(t), \dots, s_M(t)\}$. Let $\{\xi_1(t), \dots, \xi_N(t)\}$ be an orthonormal basis for the signal subspace generated by $\mathcal{S}$, i.e., for the space of all signals obtained as linear combinations of the $s_m(t)$'s. From the Gram-Schmidt theorem, we know that this basis exists (not unique) and that $N \leq M$. Then, let $\mathcal{B} = \{\xi_1(t), \dots, \xi_N(t), \xi_{N+1}(t), \dots\}$ be a complete orthonormal basis for the set of finite energy signals over $[0, T]$. As before, the discrete representation of the received signal with respect to $\mathcal{B}$ is given by the sequence $\mathbf{r} = (r_1, r_2, \dots)$ such that

$$\begin{aligned} r_1 &= s_{m,1} + \nu_1 \\ r_2 &= s_{m,2} + \nu_2 \\ &\vdots \\ r_N &= s_{m,N} + \nu_N \\ r_{N+1} &= \nu_{N+1} \\ r_{N+2} &= \nu_{N+2} \\ &\vdots \end{aligned}$$

where r_i is given by (2.3), and where $\mathbf{s}_m = (s_{m,1}, \dots, s_{m,N})$ is the vector representation of the signal $s_m(t)$ with respect to the basis $\{\xi_1(t), \dots, \xi_N(t)\}$. By construction, all signals $s_m(t)$ can be represented exactly in such way, and they are orthogonal to all the elements $\xi_i(t)$ of the basis $\mathcal{B}$ for $i > N$. Since all components r_i for $i > N$ are independent of the transmitted message m , the desired sufficient statistics is given by the vector $\mathbf{y}$ corresponding to the first N components of $\mathbf{r}$.

The resulting N -dimensional discrete channel model is given by

$$\mathbf{y} = \mathbf{s}_m + \boldsymbol{\nu} \tag{2.8}$$

where $\boldsymbol{\nu} \sim \mathcal{N}_{\mathbb{C}}(0, N_0 \mathbf{I})$.

The conditional pdf of the observation $\mathbf{y}$ given m is the N -variate complex Gaussian pdf

$$p(\mathbf{y}|m) = \frac{1}{(\pi N_0)^N} \exp\left(-\frac{1}{N_0} |\mathbf{y} - \mathbf{s}_m|^2\right) \quad (2.9)$$

In order to determine the MAP decision rule, we have to find the decision regions $\mathcal{D}_m$, for all $m = 1, \dots, M$. The region $\mathcal{D}_m$ is given by all $\mathbf{y} \in \mathbb{C}^N$ satisfying the inequalities

$$\pi(m) \frac{1}{(\pi N_0)^N} \exp\left(-\frac{1}{N_0} |\mathbf{y} - \mathbf{s}_m|^2\right) \geq \pi(m') \frac{1}{(\pi N_0)^N} \exp\left(-\frac{1}{N_0} |\mathbf{y} - \mathbf{s}_{m'}|^2\right)$$

for all $m' \neq m$. Each pair of symbols (m, m') determines one inequality, the solution of which is a *pairwise decision region* in $\mathbb{C}^N$. The desired decision region $\mathcal{D}_m$ is obtained as the intersection of all those pairwise decision regions. Let us consider the pair (m, m') . The associated pairwise decision region is given by all $\mathbf{y}$ satisfying

$$|\mathbf{y} - \mathbf{s}_m|^2 - |\mathbf{y} - \mathbf{s}_{m'}|^2 \leq N_0 \log \frac{\pi(m)}{\pi(m')} \quad (2.10)$$

It is immediate to see that the boundary of the above region is the hyperplane

$$2\operatorname{Re}\left\{(\mathbf{s}_{m'} - \mathbf{s}_m)^H \mathbf{y}\right\} = N_0 \log \frac{\pi(m)}{\pi(m')} - |\mathbf{s}_m|^2 + |\mathbf{s}_{m'}|^2 \quad (2.11)$$

i.e., the affine variety obtained by translating the subspace perpendicular to the difference vector $\mathbf{d} = \mathbf{s}_{m'} - \mathbf{s}_m$ by a quantity that depends on the difference of signal energies, on the a priori probabilities and on the noise variance.

The MAP decision rule can be written compactly in the form

$$\hat{m} = g(\mathbf{y}) = \arg \max_{m \in \{1, \dots, M\}} \left\{ \log \pi(m) - \frac{1}{N_0} |\mathbf{y} - \mathbf{s}_m|^2 \right\} \quad (2.12)$$

In the important case of uniform prior probabilities, the decision region (2.10) becomes

$$|\mathbf{y} - \mathbf{s}_m|^2 \leq |\mathbf{y} - \mathbf{s}_{m'}|^2 \quad (2.13)$$

This means that the detector chooses m instead of m' if the received vector $\mathbf{y}$ is closer to $\mathbf{s}_m$ than to $\mathbf{s}_{m'}$. The resulting ML decision rule is the intuitive *minimum squared Euclidean distance (SED) rule*

$$\hat{m} = g(\mathbf{y}) = \arg \min_{m \in \{1, \dots, M\}} \{|\mathbf{y} - \mathbf{s}_m|^2\} \quad (2.14)$$

As in the case of binary antipodal signals, we have that the detector can be divided into two distinct operations: a projection onto the signal space, in order to extract the sufficient statistic, followed by a decision based on the sufficient statistics. The components y_i of the observation $\mathbf{y}$ can be obtained either by a bank of N correlators, calculating $y_i = (\xi_i(t)|r(t))$ for $i = 1, \dots, N$, or by a bank of matched filters, with impulse responses $h_i(t) = \xi_i(T - t)^*$, whose output is sampled at time T . Figs. ?? and ?? represent the block diagrams of the optimal MAP detector for general M -ary signals based on correlation and on matched filtering, respectively.

Figure 2.4: MAP detector for M -ary signals based on a bank of signal correlators.

Figure 2.5: MAP detector for M -ary signals based on a bank of matched filters.

As an alternative, it is sometimes convenient to work directly with the signals $s_m(t)$ rather than with an orthonormal basis. By writing $|\mathbf{y} - \mathbf{s}_m|^2 = |\mathbf{y}|^2 - 2\text{Re}\{\mathbf{s}_m^H \mathbf{y}\} + |\mathbf{s}_m|^2$, we notice that the received signal energy term $|\mathbf{y}|^2$ does not play any role in the decision, since it is common to all hypotheses. Moreover, the vector inner product $\mathbf{s}_m^H \mathbf{y}$ can be written in terms of the continuous-time signals as

$$\mathbf{s}_m^H \mathbf{y} = (s_m(t)|r(t)) = \int s_m(t)^* r(t) dt$$

Therefore, the minimum distance rule (2.14) can be replaced by the maximization

$$\hat{x} = g(\mathbf{y}) = \arg \max_{m \in \{1, \dots, M\}} \left\{ 2\text{Re} \left\{ \int s_m(t)^* r(t) dt \right\} - |\mathbf{s}_m|^2 \right\} \quad (2.15)$$

The corresponding detector, implemented by a bank of M correlators (or equivalently, by a bank of M matched filters), is represented in Fig. ??.

Figure 2.6: ML detector for M -ary signals based on a bank of signal correlators.

We have seen that there are several different equivalent formulations of the same optimal detector. In general, we say that a real function $\mathcal{M}(\mathbf{y}, m)$ is a *decision metric* for the MAP detection of m given the observation $\mathbf{y}$ if the outcome of

$$\hat{m} = \arg \text{sel}_{m \in \{1, \dots, M\}} \mathcal{M}(\mathbf{y}, m)$$

where “sel” denotes either the “min” or the “max” selection operation, is the same of the MAP rule (2.12).

2.3 Performance of coherent detection

For a general signal set $\mathcal{S} = \{\mathbf{s}_m : m = 1, \dots, M\}$, the average symbol error probability (assuming uniform prior probability) is given by

$$P(e) = \Pr(\hat{m} \neq m) = \frac{1}{M} \sum_{m=1}^M \Pr(\hat{m} \neq m|m)$$

In general, the above probability is difficult if not impossible to compute in closed form. Therefore, we shall resort to the union upper bound (1.32) and to the lower bound (1.37). Next, we shall see some important particular cases for which $P(e)$ finds a closed form expression or for which it is possible to find tighter upper bounds than the simple union bound.

The average symbol error probability is upperbounded by

$$P(e) \leq \frac{1}{M} \sum_{m=1}^M \sum_{m' \neq m} P(m \rightarrow m') \quad (2.16)$$

where the PEP is given by

$$\begin{aligned} P(m \rightarrow m') &= \Pr(|\mathbf{y} - \mathbf{s}_{m'}|^2 \leq |\mathbf{y} - \mathbf{s}_m|^2 | m) \\ &= \Pr(2\text{Re}\{(\mathbf{s}_{m'} - \mathbf{s}_m)^H \boldsymbol{\nu}\} \geq |\mathbf{s}_{m'} - \mathbf{s}_m|^2) \\ &= \Pr(\eta \geq d^2(m, m')) \end{aligned} \quad (2.17)$$

where we have defined the SED between the signals $\mathbf{s}_m$ and $\mathbf{s}_{m'}$ as

$$d^2(m, m') = |\mathbf{s}_{m'} - \mathbf{s}_m|^2$$

and where η is a real Gaussian random variable $\sim \mathcal{N}(0, 2N_0 d^2(m, m'))$. Recalling the definition of the Gaussian tail function $Q(u)$, we can write

$$P(m \rightarrow m') = Q\left(\sqrt{\frac{d^2(m, m')}{2N_0}}\right) \quad (2.18)$$

The resulting general expression for the union bound on the average symbol error probability is

$$P(e) \leq \frac{1}{M} \sum_{m=1}^M \sum_{m' \neq m} Q\left(\sqrt{\frac{d^2(m, m')}{2N_0}}\right) \quad (2.19)$$

In order to provide a fair comparison of different signal sets, it is useful to express their symbol error probability as a function of the ratio E_b/N_0 , where E_b is the average energy per transmitted bit. To this purpose, we define the average energy per transmitted symbols as

$$E_s = \frac{1}{M} \sum_m |\mathbf{s}_m|^2$$

Since the symbols are transmitted with uniform probability, each symbol carries $\log_2 M$ bits of information (in order words, we can label each signal in $\mathcal{S}$ by binary labels of $\log_2 M$ bit each). Therefore, E_b is given by

$$E_b = E_s / \log_2 M$$

We can define a normalized *unit-energy* signal set $\tilde{\mathcal{S}}$ obtained from $\mathcal{S}$ by dividing all its vectors by $\sqrt{E_s}$. Obviously, for any pair of symbols (m, m') , the distance $d^2(m, m')$ is related to the corresponding distance $\tilde{d}^2(m, m')$ between signals in $\tilde{\mathcal{S}}$ by

$$d^2(m, m') = E_s \tilde{d}^2(m, m')$$

Finally, the union bound can be conveniently expressed as a function of E_b/N_0 by using the normalized distances in (2.19). We obtain

$$P(e) \leq \frac{1}{M} \sum_{m=1}^M \sum_{m' \neq m} Q \left(\sqrt{\frac{\log_2 M \tilde{d}^2(m, m')}{2} \frac{E_b}{N_0}} \right) \quad (2.20)$$

Notice that if the signal set has the *uniform distance property*, i.e., if the list of distances $\{d^2(m, m') : m' = 1, \dots, M\}$ is independent of the reference symbol m , then the outer average with respect to m is not needed, and the bound can be calculated by taking any suitable m as reference.

The Chernoff union bound to the symbol error probability is obtained by noticing that the LLR $L_{m,m'}(\mathbf{y}) = \log p(\mathbf{y}|\mathbf{s}_m)/p(\mathbf{y}|\mathbf{s}_{m'})$ is given by

$$\begin{aligned} L_{m,m'}(\mathbf{y}) &= \frac{1}{N_0} (-|\mathbf{y} - \mathbf{s}_m|^2 + |\mathbf{y} - \mathbf{s}_{m'}|^2) \\ &= \frac{1}{N_0} (-|\boldsymbol{\nu}|^2 + |\boldsymbol{\nu}|^2 - 2\text{Re}\{(\mathbf{s}_{m'} - \mathbf{s}_m)^H \boldsymbol{\nu}\} + d^2(m, m')) \\ &= \frac{1}{N_0} (-2\text{Re}\{(\mathbf{s}_{m'} - \mathbf{s}_m)^H \boldsymbol{\nu}\} + d^2(m, m')) \end{aligned} \quad (2.21)$$

i.e., it is a Gaussian real random variable $\sim \mathcal{N}(d^2(m, m')/N_0, 2d^2(m, m')/N_0)$. Its characteristic function on the real positive axis is given by

$$\Phi_L(\lambda) = \exp \left((\lambda^2 - \lambda) \frac{d^2(m, m')}{N_0} \right)$$

By minimizing this with respect to λ , we get that the minimum is achieved by $\lambda = 1/2$ and it is given by

$$\exp \left(-\frac{d^2(m, m')}{4N_0} \right)$$

Finally, the resulting union Chernoff bound is given by

$$P(e) \leq \frac{1}{M} \sum_{m=1}^M \sum_{m' \neq m} \exp \left(-\frac{\log_2 M \tilde{d}^2(m, m')}{4} \frac{E_b}{N_0} \right) \quad (2.22)$$

We notice that this upper bound could have been obtained directly by applying the upper bound

$$Q(u) \leq \exp(-u^2/2)$$

(valid for $u \geq 0$) to the union bound (2.20).

From the general lower bound (1.37), we obtain a lower bound on the symbol error probability as

$$P(e) \geq \frac{1}{M} \sum_{m=1}^M \max_{m' \neq m} Q \left(\sqrt{\frac{\log_2 M \tilde{d}^2(m, m')}{2} \frac{E_b}{N_0}} \right) \quad (2.23)$$

Since $Q(u)$ is monotonically decreasing, the maximum over all $m' \neq m$ of $Q \left(\sqrt{\frac{\log_2 M \tilde{d}^2(m, m')}{2} \frac{E_b}{N_0}} \right)$ is obtained for the signal $\mathbf{s}_{m'}$ at minimum distance from $\mathbf{s}_m$. Therefore, we can also write the lower bound as

$$P(e) \geq \frac{1}{M} \sum_{m=1}^M Q \left(\sqrt{\frac{\log_2 M \min_{m' \neq m} \tilde{d}^2(m, m')}{2} \frac{E_b}{N_0}} \right) \quad (2.24)$$

The same observation regarding the uniform distance property done for the union upper bound applies here.

2.3.1 Simplifications and importance of the minimum SED

The upper and lower bounds (2.20) and (2.24) require the evaluation of all distances between any pair of signals $\mathbf{s}_m, \mathbf{s}_{m'}$. Looser upper and lower bounds can be obtained in terms of the minimum distance of the normalized signal set, defined by

$$d_{\min}^2(\mathcal{S}) = \min_{m \neq m'} \tilde{d}^2(m, m') \quad (2.25)$$

Because of the monotonicity of $Q(u)$, each term in (2.20) is upperbounded by substituting $d_{\min}^2(\mathcal{S})$ to the corresponding distance $\tilde{d}^2(m, m')$. Then, we obtain the bound

$$P(e) \leq (M-1)Q \left(\sqrt{\frac{\log_2 M d_{\min}^2(\mathcal{S})}{2} \frac{E_b}{N_0}} \right) \quad (2.26)$$

In order to simplify the lower bound, we can trivially lower bound each term in (2.24) for which $\min_{m' \neq m} \tilde{d}^2(m, m') > d_{\min}^2(\mathcal{S})$ by zero. The resulting looser lower bound is

$$P(e) \geq f_{\min} Q \left(\sqrt{\frac{\log_2 M d_{\min}^2(\mathcal{S})}{2} \frac{E_b}{N_0}} \right) \quad (2.27)$$

where $0 < f_{\min} < 1$ is the fraction of signals having at least a *nearest neighbor* at distance $d_{\min}^2(\mathcal{S})$.

Finally, we can obtain an approximation of $P(e)$ (neither an upper nor a lower bound) by simply eliminating from the union bound all terms with distance larger than the minimum distance. We obtain

$$P(e) \approx \bar{N}_{\min} Q \left(\sqrt{\frac{\log_2 M d_{\min}^2(\mathcal{S})}{2} \frac{E_b}{N_0}} \right) \quad (2.28)$$

where $\bar{N}_{\min}$ is the average number of nearest neighbors at minimum distance, given by

$$\bar{N}_{\min} = \frac{1}{M} \sum_{m=1}^M N_{\min}(m)$$

where $N_{\min}(m)$ is the number of nearest neighbors at distance $d_{\min}^2(\mathcal{S})$ for the signal $\mathbf{s}_m$.

We observe that, both the upper and lower bound on $P(e)$ (as well as the above approximation) have the same exponential behavior with respect to E_b/N_0 . We conclude that $P(e) = O(e^{-0.25\gamma(\mathcal{S})E_b/N_0})$ where the coefficient in the exponent is given by

$$\gamma(\mathcal{S}) = \log_2 M d_{\min}^2(\mathcal{S}) \quad (2.29)$$

(recall that the minimum distance is defined for a signal set with normalized unit average energy).

The *asymptotic gain* of a signal set $\mathcal{S}$ over another signal set $\mathcal{S}'$ is defined by the ratio, expressed in dB, of the coefficients $\gamma(\mathcal{S})/\gamma(\mathcal{S}')$. This corresponds, for small values of $P(e)$, to the increment of E_b/N_0 needed by $\mathcal{S}'$ in order to have the same symbol error probability of $\mathcal{S}$ (notice that this increment, or “gain”, can be negative).

2.4 Signal sets for coherent detection

In this section, the most common signal sets used in digital communications systems based on coherent detection are presented and their error probability is calculated or approximated. In many cases, by exploiting the particular geometry of the signal sets, it is possible to find exact error probability expressions and upper bounds tighter than the general union bound.

In general, a signal set $\mathcal{S} = \{s_m(t) : m \in \{1, \dots, M\}\}$ where $s_m(t)$ are all proportional to a single basic waveform $\psi(t)$, i.e., they are obtained by modulating the amplitude and the phase of a basic waveform as

$$s_m(t) = a_m \psi(t) \quad (2.30)$$

via complex amplitudes $a_m \in \mathbb{C}$, is called (somewhat improperly) a “linear modulation scheme”.

In this case, the signal space has dimension one, and it is generated by the vector $\xi_1(t) = \psi(t)/\|\psi(t)\|$. Without loss of generality, we assume that

$$\frac{1}{M} \sum_{m=1}^M |a_m|^2 = 1$$

so that $E_s = \|\psi(t)\|^2$ is the average symbol energy. Signals in $\mathcal{S}$ are represented as points in the complex plane (one-dimensional complex vectors)

$$s_m = \sqrt{E_s} a_m$$

As usual, we denote by $\mathcal{S}$ both the set of signals and the corresponding set of points, that in this case is also referred to as *signal constellation*.

The most common signal sets of this type are pulse-amplitude modulation (PAM), quadrature amplitude modulation (QAM) and phase shift keying (PSK).

2.4.1 Pulse-Amplitude Modulation

The signal constellation for M -PAM is formed by M points on the real axis. Usually, these points are equally spaced and symmetrically placed with respect to the origin (antipodal PAM). In this case, we have

$$\mathcal{S} = \{\Delta(2m - M + 1) : m = 0, \dots, M - 1\}$$

where Δ is a real parameter. The relation between Δ and E_s is given by

$$\begin{aligned} E_s &= \frac{\Delta^2}{M} \sum_{m=0}^{M-1} (2m - M + 1)^2 \\ &= \Delta^2 \frac{M^2 - 1}{3} \end{aligned} \quad (2.31)$$

and the constellation minimum SED is $d_{\min}^2(\mathcal{S}) = 4\Delta^2 = 12 \log_2 M E_b / (M^2 - 1)$. Fig. ?? shows an example for the 4-PAM constellation.

Figure 2.7: 4-PAM signal constellation.

The symbol error probability of M -PAM is easily upperbounded by

$$P(e) \leq 2Q \left(\sqrt{\frac{6 \log_2 M E_b}{M^2 - 1 N_0}} \right) \quad (2.32)$$

since every signal has at most two other signals at minimum distance, and since all pairwise error regions between signals at distance larger than the minimum distance are included into regions of signals at minimum distance.

We can find an exact expression for $P(e)$ as follows. We notice that the decision regions for all signals are congruent, but for the two extremal signals. Then, we can write,

$$P(e) = \frac{1}{M} \left[\sum_{m=1}^{M-2} \Pr(\mathbf{y} \notin \mathcal{D}_m | m) + \Pr(\mathbf{y} \notin \mathcal{D}_0 | m = 0) + \Pr(\mathbf{y} \notin \mathcal{D}_{M-1} | m = M - 1) \right]$$

By symmetry, we see that all inner points have the same conditional error probability and the two outer points have the same conditional error probability (we need to compute only two terms, and not M). For the inner points we have

$$\begin{aligned}\Pr(\mathbf{y} \notin \mathcal{D}_m|m) &= \Pr(\{\text{Re}\{\boldsymbol{\nu}\} < -\Delta\} \cup \{\text{Re}\{\boldsymbol{\nu}\} > \Delta\}) \\ &= 2Q(\sqrt{2\Delta^2/N_0})\end{aligned}$$

For the outer points we have

$$\begin{aligned}\Pr(\mathbf{y} \notin \mathcal{D}_0|m=0) &= \Pr(\text{Re}\{\boldsymbol{\nu}\} > \Delta) \\ &= Q(\sqrt{2\Delta^2/N_0})\end{aligned}$$

Finally, we obtain

$$\begin{aligned}P(e) &= \frac{2(M-1)}{M}Q(\sqrt{2\Delta^2/N_0}) \\ &= \frac{2(M-1)}{M}Q\left(\sqrt{\frac{6\log_2 M E_b}{M^2-1 N_0}}\right)\end{aligned}\tag{2.33}$$

We observe that the factor 2 of the upper bound (2.32) is changed into the factor $2(M-1)/M$ in the true error probability expression.

2.4.2 Quadrature-Amplitude Modulation

The signal constellation for M -QAM is formed by M points on the complex plane. Usually, these points belong to a square lattice and are symmetrically placed with respect to the origin (squared QAM). When M is a square, the squared M -QAM constellation can be seen as two $\sqrt{M}$ -PAM constellation in phase quadrature, with points of coordinates

$$\mathcal{S} = \{\Delta[(2m - M' + 1) + j(2n - M' + 1)] : m = 0, \dots, M' - 1, n = 0, \dots, M' - 1\}$$

where $M' = \sqrt{M}$ and where Δ is a real parameter. The relation between Δ and E_s is given by

$$\begin{aligned}E_s &= \frac{\Delta^2}{M} \sum_{m=0}^{M'-1} \sum_{n=0}^{M'-1} (2m - M' + 1)^2 + (2n - M' + 1)^2 \\ &= 2\Delta^2 \frac{M-1}{3}\end{aligned}\tag{2.34}$$

and the constellation minimum SED is $d_{\min}^2(\mathcal{S}) = 4\Delta^2 = 6\log_2 M E_b / (M-1)$. Fig. ?? shows an example for the 16-QAM constellation.

In general, we can upperbound the symbol error probability of squared QAM by noticing that the decision region of each constellation point is larger or equal to a square of side 2Δ ,

Figure 2.8: 16-QAM signal constellation.

centered in the point. Then, we have

$$\begin{aligned}
 P(e) &\leq 1 - \left(1 - 2Q \left(\sqrt{\frac{2\Delta^2}{N_0}} \right) \right)^2 \\
 &= 4Q \left(\sqrt{\frac{2\Delta^2}{N_0}} \right) \left(1 - Q \left(\sqrt{\frac{2\Delta^2}{N_0}} \right) \right) \\
 &= 4Q \left(\sqrt{\frac{3 \log_2 M}{M-1} \frac{E_b}{N_0}} \right) \left(1 - Q \left(\sqrt{\frac{3 \log_2 M}{M-1} \frac{E_b}{N_0}} \right) \right) \quad (2.35)
 \end{aligned}$$

In the case where M is a square, $P(e)$ can be calculated exactly by noticing that a symbol error occurs either when the in-phase (I) or when the quadrature (Q) symbol components are erroneously detected. Since the real and imaginary parts of the noise are i.i.d. and the decision regions have boundaries parallel to the real and imaginary axes, detection errors in the I and Q components are statistically independent. Then, the exact probability of error can be written as

$$P(e) = 1 - (1 - P_{\text{PAM}}(e))^2$$

where $P_{\text{PAM}}(e)$ is the symbol error probability of the corresponding $\sqrt{M}$ -PAM in the I and Q components. By using (2.33) we obtain

$$P(e) = 4 \frac{\sqrt{M}-1}{\sqrt{M}} Q \left(\sqrt{\left(\frac{3 \log_2 M}{M-1} \right) \frac{E_b}{N_0}} \right) \left(1 - \frac{\sqrt{M}-1}{\sqrt{M}} Q \left(\sqrt{\left(\frac{3 \log_2 M}{M-1} \right) \frac{E_b}{N_0}} \right) \right) \quad (2.36)$$

2.4.3 Phase-Shift Keying

In M -PSK, the information messages are encoded in the phase of the signal. The corresponding signal constellation is formed by M points equally spaced on a circle in the complex plane centered in the origin, with radius Δ , i.e.,

$$\mathcal{S} = \{\Delta \exp(j2\pi m/M) : m = 0, \dots, M-1\}$$

Notice that 2-PSK (also denoted by BPSK) coincides with antipodal 2-PAM, and 4-PSK (also denoted by QPSK) coincides with 4-QAM. The average energy of M -PSK is given simply by $E_s = \Delta^2$, and the minimum distance is $d_{\min}^2(\mathcal{S}) = 4\Delta^2 \sin^2(\pi/M) = 4 \log_2 M \sin^2(\pi/M) E_b$. Fig. ?? shows an example for the 8-PSK constellation.

Figure 2.9: 8-PSK signal constellation.

The symbol error probability for BPSK and QPSK is given exactly by using (2.33) and (2.36) for $M = 2$ and for $M = 4$, respectively. For $M > 4$ it is not possible to find an exact expression of $P(e)$. However, a tight upper bound is obtained by noticing that the error region for any reference symbol is included in the union of the two pairwise regions given by the two symbols at minimum distance with respect to the reference symbol. This implies that the union bound can be limited to only two terms at minimum distance. The resulting upper bound is given by

$$P(e) \leq 2Q \left(\sqrt{2 \log_2 M \sin^2(\pi/M) \frac{E_b}{N_0}} \right) \quad (2.37)$$

2.5 Non-coherent detection

We assume that the modulation waveforms belong to an N -dimensional signal set $\mathcal{S}$ of size M , generated by an orthonormal basis $\mathcal{B} = \{\xi_1(t), \dots, \xi_N(t)\}$. With respect to the basis $\mathcal{B}$, the modulation signals are represented as N -dimensional complex vectors $\mathbf{s}_m = (s_{m,1}, \dots, s_{m,N})^T$, so that

$$s_m(t) = \sum_{i=1}^N s_{m,i} \xi_i(t) \quad (2.38)$$

The received signal is given by (2.1), i.e., $y(t) = s_m(t)e^{j\theta} + \nu(t)$, for $t \in [0, T]$. By projecting $y(t)$ onto the versor $\xi_i(t)$ and letting $\mathbf{y} = (y_1, \dots, y_N)^T$ be the projected vector, we obtain the vector channel model

$$\mathbf{y} = \mathbf{s}_m e^{j\theta} + \boldsymbol{\nu} \quad (2.39)$$

where $\boldsymbol{\nu} \sim \mathcal{N}_c(\mathbf{0}, N_0 \mathbf{I})$. As already said at the beginning of this chapter, a non-coherent receiver does not estimate explicitly the carrier phase and treats it as random variables, uniformly distributed over $[-\pi, \pi)$.

The signal vector $\mathbf{y}$ in (2.39) is Gaussian conditionally on the transmitted symbol m and on the phase θ . Its conditional pdf is given by

$$p(\mathbf{y}|m, \theta) = \frac{1}{(\pi N_0)^N} \exp\left(-\frac{1}{N_0} |\mathbf{y} - \mathbf{s}_m e^{j\theta}|^2\right) \quad (2.40)$$

The non-coherent ML decision rule for the detection symbol m with observation $\mathbf{y}$ is given by

$$\hat{m} = \arg \max_{m \in \{1, \dots, M\}} p(\mathbf{y}|m) \quad (2.41)$$

where $p(\mathbf{y}|m)$ is obtained from $p(\mathbf{y}|m, \theta)$ by averaging with respect to the phase θ .

We have

$$\begin{aligned} p(\mathbf{y}|m) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} p(\mathbf{y}|m, \theta) d\theta \\ &= \frac{1}{(\pi N_0)^N} \exp\left(-\frac{|\mathbf{y}|^2 + |\mathbf{s}_m|^2}{N_0}\right) \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp\left(\frac{2}{N_0} \operatorname{Re}\{\mathbf{y}^H \mathbf{s}_m e^{j\theta}\}\right) d\theta \\ &= \frac{1}{(\pi N_0)^N} \exp\left(-\frac{|\mathbf{y}|^2 + |\mathbf{s}_m|^2}{N_0}\right) \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp\left(\frac{2}{N_0} |\mathbf{y}^H \mathbf{s}_m| \cos(\theta + \phi)\right) d\theta \\ &= \frac{1}{(\pi N_0)^N} \exp\left(-\frac{|\mathbf{y}|^2 + |\mathbf{s}_m|^2}{N_0}\right) I_0\left(\frac{2}{N_0} |\mathbf{y}^H \mathbf{s}_m|\right) \end{aligned} \quad (2.42)$$

where we used the integral definition of the modified Bessel function [?] $I_0(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp(z \cos \theta) d\theta$ and we observed that if θ is uniformly distributed over $[-\pi, \pi)$, also $\theta + \phi$ is uniformly distributed over the same interval, where $\phi = \angle \mathbf{y}^H \mathbf{s}_m$ is a phase independent of θ .

The maximization of the above likelihood function for a general signal set $\mathcal{S}$ does not lead to particularly appealing structures. However, in the special case of equal energy signals (i.e., signal sets $\mathcal{S}$ where $|\mathbf{s}_m|^2 = E_s$ for all $m = 1, \dots, M$), we notice that $p(\mathbf{y}|m)$ is an increasing function of the squared magnitude of the correlation $|\mathbf{s}_m^H \mathbf{y}|^2$. From now on, we make the assumption of equal energy signals and we rewrite the ML decision rule as

$$\hat{m} = \arg \max_{m \in \{1, \dots, M\}} |\mathbf{s}_m^H \mathbf{y}|^2 \quad (2.43)$$

This rule is called *square-law detector* and can be implemented by a bank of correlators, which correlate the received signal vector $\mathbf{y}$ with all possible transmitted signal vectors $\mathbf{s}_m \in \mathcal{S}$, followed by a comparator that selects the signal whose correlation has maximum squared magnitude (squared envelope). Obviously, the squared magnitude of the signal correlation does not depend any longer on the signal phase, therefore, this ML detector is insensitive to the phase rotation introduced by the channel. Fig. ?? shows the structure of a non-coherent ML detector.

Figure 2.10: Block diagram of a non-coherent ML detector for equal energy signals.

2.6 Performance of non-coherent detection

In this section we consider only equal-energy signals, for which the ML non-coherent detector coincides with the squared-envelope detector. As for the case of coherent detection, we compute the decision metric difference between the correct signal $\mathbf{s}_m$ and a wrong signal $\mathbf{s}_{m'}$. Without loss of generality, we let $\theta = 0$ since the receiver is independent of the carrier phase. Then,

$$\begin{aligned}
 \Delta &= |\mathbf{s}_m^H \mathbf{y}|^2 - |\mathbf{s}_{m'}^H \mathbf{y}|^2 \\
 &= |\mathbf{s}_m^H (\mathbf{s}_m + \boldsymbol{\nu})|^2 - |\mathbf{s}_{m'}^H (\mathbf{s}_m + \boldsymbol{\nu})|^2 \\
 &= |E_s + v|^2 - |\rho E_s + v'|^2 \\
 &= \mathbf{z}^H \mathbf{F} \mathbf{z}
 \end{aligned} \tag{2.44}$$

where we define the proper Gaussian vector $\mathbf{z} = (E_s + v, \rho E_s + v')^T$, with $v = \mathbf{s}_m^H \boldsymbol{\nu}$, $v' = \mathbf{s}_{m'}^H \boldsymbol{\nu}$, the signal energy $E_s = |\mathbf{s}_m|^2 = |\mathbf{s}_{m'}|^2$, the signal correlation coefficient $\rho = \frac{1}{E_s} \mathbf{s}_{m'}^H \mathbf{s}_m$ and the

Hermitian 2×2 matrix

$$\mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (2.45)$$

The vector $\mathbf{z}$ has mean $\bar{\mathbf{z}} = (E_s, \rho E_s)^T$ and covariance matrix

$$\mathbf{R} = N_0 E_s \begin{bmatrix} 1 & \rho^* \\ \rho & 1 \end{bmatrix}$$

The PEP is given by $P(m \rightarrow m') = P(\Delta \leq 0)$. The pdf of Δ cannot be found in closed form. However, we are able to find the characteristic function of Δ , defined by

$$\Phi_\Delta(s) = E[\exp(-s\Delta)] = \int \exp(-sz) p_\Delta(z) dz$$

Then, the PEP can be evaluated by the Laplace transform inversion formula

$$P(\Delta \leq 0) = \frac{1}{j2\pi} \int_{\sigma-j\infty}^{\sigma+j\infty} \frac{1}{s} \Phi_\Delta(s) ds \quad (2.46)$$

where the vertical integration path $s = \sigma + j\omega$, $\omega \in \mathbb{R}$ must belong to the region of convergence (ROC) of $\Phi_\Delta(s)/s$. The ROC is the vertical strip $0 < \sigma < \alpha_+$, where α_+ is the smallest positive real part of singularities of $\Phi_\Delta(s)$ (see [?, ?, ?] and references therein). Several efficient numerical techniques are available for Laplace inversion [?] and, more in general, for the evaluation of $P(\Delta \leq 0)$ from the characteristic function $\Phi_\Delta(s)$. A simpler upper bound on the PEP, that often yields enough qualitative information, is the Chernoff bound

$$P(\Delta \leq 0) \leq \min_{0 < \lambda < \alpha_+} \Phi_\Delta(\lambda) \quad (2.47)$$

In our case, (2.44) turns out to be a *Hermitian Quadratic Form of complex Gaussian Random Variables* (briefly, HQF-GRV), for which a general expression of the characteristic function is known [?]. The following result is proved in Appendix 2.9

Proposition. The characteristic function of the HQF-GRV $\Delta = \mathbf{z}^H \mathbf{F} \mathbf{z}$, where $\mathbf{z} \sim \mathcal{N}_{\mathbb{C}}(\bar{\mathbf{z}}, \mathbf{R})$ is given by

$$\Phi_\Delta(s) = \frac{\exp(-s\bar{\mathbf{z}}^H \mathbf{F} (\mathbf{I} + s\mathbf{R}\mathbf{F})^{-1} \bar{\mathbf{z}})}{\det(\mathbf{I} + s\mathbf{R}\mathbf{F})} \quad (2.48)$$

□

After some algebra and the change of variable $(1 - |\rho|^2)E_s^2 s \rightarrow s$, we obtain

$$\Phi_\Delta(s) = \frac{\exp\left(-\frac{s(1-a)}{1-bs^2}\right)}{1-bs^2} \quad (2.49)$$

where the coefficients a and b are given by

$$\begin{aligned} a &= \frac{N_0}{E_s} \frac{1}{1-|\rho|^2} \\ b &= \left(\frac{N_0}{E_s}\right)^2 \frac{1}{1-|\rho|^2} \end{aligned}$$

In the case where $\Phi_\Delta(s)$ has only poles (i.e., no essential singularities), we can use the Laplace inversion formula and compute exactly the PEP by using the *residue theorem* [?]:

$$\frac{1}{j2\pi} \int_{\sigma-j\infty}^{\sigma+j\infty} \frac{1}{s} \Phi_\Delta(s) ds = \begin{cases} -\sum_{s_i \in \text{RHP poles}} \text{Res}(\Phi_\Delta(s)/s, s_i) \\ 1 + \sum_{s_i \in \text{LHP poles}} \text{Res}(\Phi_\Delta(s)/s, s_i) \end{cases} \quad (2.50)$$

where RHP and LHP denote the right half-plane ($\text{Re}\{s\} > 0$) and left half-plane ($\text{Re}\{s\} < 0$) of the complex plane, and where $\text{Res}(f(s), s_i)$ denotes the residue of a function $f(s)$ in s_i . For poles of multiplicity k , we have the following lemma:

Lemma: residue formula. Let $f(s) = g(s)/(s - s_0)^k$ with $g(s)$ analytic in s_0 . Then, the residue of $f(s)$ in $s = s_0$ is given by

$$\text{Res}(f(s), s_0) = \frac{1}{(k-1)!} \left. \frac{d^{k-1}}{ds^{k-1}} g(s) \right|_{s=s_0}$$

□

For example, in the case of orthogonal signals ($\rho = 0$), we obtain the PEP in closed form. In fact, by letting $\rho = 0$ in (2.49) we obtain

$$\Phi_\Delta(s) = \frac{\exp(-s/(1 + sN_0/E_s))}{(1 + sN_0/E_s)(1 - sN_0/E_s)}$$

This function has two poles of order 1, located in $\pm E_s/N_0$. By applying the residue formula we obtain

$$P(m \rightarrow m') = \frac{1}{2} e^{-\frac{1}{2} \frac{E_s}{N_0}} \quad (2.51)$$

With some more involved calculation (see [3]), it is possible to obtain a general result for non-orthogonal equal energy signals in the following form

$$P(m \rightarrow m') = Q(u, v) - \frac{1}{2} e^{-(u^2+v^2)/2} I_0(uv) \quad (2.52)$$

where we define

$$\begin{aligned} u &= \sqrt{\frac{E_s}{2N_0}} (1 - \sqrt{1 - |\rho|^2}) \\ v &= \sqrt{\frac{E_s}{2N_0}} (1 + \sqrt{1 - |\rho|^2}) \end{aligned}$$

and where $Q(u, v)$ is the Marcum “ Q ” function, defined by the integral

$$Q(u, v) = \int_v^\infty z e^{-(z^2+u^2)/2} I_0(uz) dz \quad (2.53)$$

Notice that for $\rho = 0$ we have $u = 0$ and $v = \sqrt{E_s/N_0}$, $Q(0, v) = e^{-v^2/2}$, therefore we find again that for orthogonal signals $P(m \rightarrow m') = \frac{1}{2} e^{-E_s/(2N_0)}$, as found above in a direct way.

2.6.1 Performance of M -ary orthogonal signals

Orthogonal M -ary signals deserve a special treatment because it is possible to derive directly the *exact* error probability without resorting to the PEP and union bound. This section is dedicated to the calculation of error probability of M -ary orthogonal signals.

Let $\mathbf{s}_1, \dots, \mathbf{s}_M$ be M orthogonal N -dimensional signals, and assume, without loss of generality, that $\mathbf{s}_1$ is transmitted, so that the received signal is given by

$$\mathbf{y} = \mathbf{s}_1 + \boldsymbol{\nu}$$

(without loss of generality, we can consider $\theta = 0$ since the receiver does not depend on θ). Let $u_i = |\mathbf{s}_i^H \mathbf{y}|$ be the magnitude of the output of the i -th correlator, then

$$u_1 = |E_s + v_1|, \quad u_i = |v_i|, \quad \text{for } i = 2, \dots, M$$

where $v_i = \mathbf{s}_i^H \boldsymbol{\nu}$ is the projection of the noise on the i -th signal vector. The noise variables v_i are i.i.d. $\sim \mathcal{N}_{\mathbb{C}}(0, N_0 E_s)$. It follows that u_1 has a Rice distribution, while the u_i 's for $i = 2, \dots, M$ have a Rayleigh distribution, given by:

$$\begin{aligned} u_1 &\sim p_{u_1}(z) = \frac{z}{2E_s N_0} \exp\left(-\frac{z^2 + 4E_s^2}{4E_s N_0}\right) I_0(z/N_0) \\ u_i &\sim p_{u_i}(z) = \frac{z}{2E_s N_0} \exp\left(-\frac{z^2}{4E_s N_0}\right) \end{aligned}$$

The probability that $\mathbf{s}_1$ is correctly detected, conditionally on $u_1 = z$, can be written as

$$P(\text{correct} | u_1 = z) = P(u_2 \leq z, u_3 \leq z, \dots, u_M \leq z) = \prod_{i=2}^M P(u_i \leq z)$$

The i -th term in the product is evaluated from the Rayleigh pdf

$$P(u_i \leq z) = \int_0^z p_{u_i}(z') dz' = 1 - \exp\left(-\frac{z^2}{4E_s N_0}\right)$$

The product can be expanded by using Newton binomial formula

$$\begin{aligned} \prod_{i=2}^M P(u_i \leq z) &= \left(1 - e^{-\frac{z^2}{4E_s N_0}}\right)^{M-1} \\ &= \sum_{m=0}^{M-1} \binom{M-1}{m} (-1)^m e^{-\frac{m z^2}{4E_s N_0}} \end{aligned}$$

Finally, we average the above expression with respect to the Rician distribution of u_1 and we obtain

$$P(\text{correct}) = \sum_{m=0}^{M-1} \binom{M-1}{m} \frac{(-1)^m}{m+1} e^{-\frac{m}{m+1} E_s / N_0}$$

where we have used the result

$$\begin{aligned} \int_0^\infty \frac{z}{2ab} e^{-\frac{(m+1)z^2+4a^2}{4ab}} I_0\left(\frac{z}{b}\right) dz = \\ \frac{1}{m+1} e^{-\frac{m}{m+1} \frac{a}{b}} \int_0^\infty \frac{y}{2ab} e^{-\frac{y^2+4a^2/(m+1)}{4ab}} I_0\left(\frac{y}{b\sqrt{m+1}}\right) dy = \\ \frac{1}{m+1} e^{-\frac{m}{m+1} \frac{a}{b}} \end{aligned}$$

obtained by the change of integration variable $y = \sqrt{m+1}z$.

The exact error probability is given by $P(e) = 1 - P(\text{correct})$, that is

$$P(e) = \sum_{m=1}^{M-1} \binom{M-1}{m} \frac{(-1)^{m+1}}{m+1} e^{-\frac{m}{m+1} E_s/N_0} \quad (2.54)$$

Again, by letting $M = 2$ we obtain the exact error probability with orthogonal binary signals as $P(e) = \frac{1}{2} e^{-E_s/(2N_0)}$.

2.7 Signal sets for non-coherent detection

In this section we present some examples of equal-energy signal sets suited to non-coherent detection.

2.7.1 Frequency-Shift Keying

Among the signals suited to non-coherent detection a prominent role is played by frequency-shift keying (FSK). In FSK, information is encoded in the signal frequency. Typically, an FSK modulator can be seen as the concatenation of a PAM modulator with a *Voltage-Controlled Oscillator* (VCO), with central frequency f_0 . In the time domain, M -ary FSK signals are given by

$$s_m(t) = A \exp(j2\pi a_m t) p_T(t)$$

where, for $m = 0, \dots, M-1$,

$$a_m = \Delta(2m - M + 1)/2$$

and where $p_T(t)$ is a rectangular pulse of duration T and amplitude 1. The spacing between frequencies corresponding to different symbols is Δ . Since all signals have the same energy, the average energy per symbol is given by

$$E_s = \int_0^T |s_m(t)|^2 dt = A^2 T$$

The distance properties and error probability of FSK depend on the normalized frequency spacing ΔT . In general, the distance between two signals can be better calculated in the time domain and it is given by

$$\begin{aligned} d^2(m, m') &= A^2 \int_0^T |\exp(j2\pi a_m t) - \exp(j2\pi a_{m'} t)|^2 dt \\ &= 2E_s(1 - \text{Re}\{\rho(m, m')\}) \end{aligned} \quad (2.55)$$

where we define the correlation coefficient

$$\begin{aligned}\rho(m, m') &= \frac{1}{T} \int_0^T \exp(j2\pi\Delta(m - m')t) dt \\ &= \exp(j\pi(m - m')\Delta T) \text{sinc}((m - m')\Delta T)\end{aligned}\quad (2.56)$$

The two signals are said to be *coherently orthogonal* if $\text{Re}\{\rho(m, m')\} = 0$, and *non-coherently orthogonal* if $\rho(m, m') = 0$. From (2.56) and from the fact that the minimum (in absolute value) of the difference $m - m'$ is 1, we obtain immediately that the condition for coherent orthogonality is that $\Delta T = \frac{k}{2}$ for $k \in \mathbb{Z}$, and the condition for non-coherent orthogonality is that $\Delta T = k$ for $k \in \mathbb{Z}$.

2.7.2 Hadamard orthogonal codes

We can obtain an orthogonal signal set from any unitary $N \times N$ matrix $\mathbf{U}$. We define the signals

$$s_m(t) = \sum_{i=1}^N u_{m,i} \psi(t - (i - 1)T/N)$$

where $\psi(t)$ is a rectangular pulse of energy E_s and support $[0, T/N]$, and where $u_{m,i}$ are the elements of $\mathbf{U}$. Since the above are linearly modulated signals, an optimal receiver front-end is obtained by taking N samples at rate N/T at the output of a matched filter with impulse response $\sqrt{\frac{1}{E_s}} \psi(-t)^*$. Equivalently, the orthonormal basis $\mathcal{B}$ for this signal set is formed by the vectors

$$\xi_i(t) = \sqrt{\frac{1}{E_s}} \psi(t - (i - 1)T/N)$$

for $i = 1, \dots, N$. The resulting vector channel has the form (2.39) where $\mathbf{s}_m = \sqrt{E_s}(u_{m,1}, \dots, u_{m,N})^T$. In particular, if the unitary matrix is given by $\mathbf{U} = \frac{1}{\sqrt{N}}\mathbf{H}$, where $\mathbf{H}$ is a Hadamard matrix, this signal set is said to be a *Hadamard orthogonal code*. Hadamard matrices are $N \times N$ matrices with elements ± 1 and mutually orthogonal row and columns. If N is a power of 2, the $N \times N$ Hadamard matrix is said of the *Sylvester type* and can be constructed by the following recursion

$$\mathbf{H}_n = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes \mathbf{H}_{n-1}$$

initialized by $\mathbf{H}_0 = 1$.¹

2.7.3 Differential M-PSK

In the presence of a random phase rotation θ , the receiver is not able to detect information symbols encoded in the signal phase. Therefore, M -PSK with symbol-by-symbol detection is

¹ $\otimes$ denotes the Kronecker product, given by

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2n}\mathbf{B} \\ \vdots & & & \vdots \\ a_{m1}\mathbf{B} & a_{m2}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}$$

useless in a non-coherent system. In order to use M -PSK with non-coherent detection, we need to detect blocks of consecutive symbols. In particular, consider a simple scheme that transmits blocks of PSK symbols $\mathbf{x} = \sqrt{E_s}(1, x_1, x_2, \dots, x_{N-1})^T$ (with $|x_i| = 1$). The first symbol is always set to 1, in order to create a phase reference for the receiver. Notice that in this way, none of the blocks $\mathbf{x}$ can be obtained from another by a phase rotation. Therefore, the signal blocks are distinguishable even without knowledge of the phase θ .

This scheme copes with the phase ambiguity of the non-coherent receiver but wastes one out of N symbols. In order to avoid this wasting, *differential encoding* can be used [3]. The PSK information symbols $a_1, a_2, \dots, a_{N-1}$ are encoded into the PSK channel symbols $x_1, x_2, \dots, x_{N-1}$ via the recursive mapping

$$x_i = x_{i-1}a_i \quad (2.57)$$

and where x_0 is set equal to the last symbol of the previously transmitted block. This method is called “differential” since the phase of a_i determines the phase shift (difference) between x_{i-1} and x_i . It is clear that even if the sequence of symbols x_i rotated by $e^{j\theta}$ the information symbols a_i can still be recovered (in the absence of noise) by

$$a_i = \left(x_{i-1}e^{j\theta}\right)^* x_i e^{j\theta} = x_{i-1}^* x_i$$

Also, no explicit phase reference symbol is needed, since the phase reference is provided by the last symbol of the previous block.

Generalizing the idea of differential PSK to blocks of length N , we have that the sequence of N -dimensional blocks

$$(\dots, \underbrace{(1, a_{1,n}, a_{2,n}, \dots, a_{N-1,n})}_N, \underbrace{(1, a_{1,n+1}, a_{2,n+1}, \dots, a_{N-1,n+1})}_N, \dots)$$

is transmitted as

$$(\dots, \underbrace{(x_{1,n}, x_{2,n}, \dots, x_{N-1,n})}_{N-1}, \underbrace{(x_{1,n+1}, x_{2,n+1}, \dots, x_{N-1,n+1})}_{N-1}, \dots)$$

where $x_{i,n} = a_{i,n}x_{N-1,n-1}$. For detection, the n -th block of $N-1$ symbols is considered together with the leading additional symbol, in order to carry out non-coherent detection over a block of N components. All signals in the n -th block are rotated by $x_{N-1,n-1}$, and since detection is non-coherent, this rotation is irrelevant, i.e., detection must be carry out by considering the original blocks of length N with the leading symbol equal to 1. Therefore, the corresponding signal block in the vector channel model (2.39) is given by $\mathbf{x} = \sqrt{E_s}(1, x_1, \dots, x_{N-1})^T$, even though the number of PSK symbols actually sent on the channel is only $N-1$ per block (apart from the very first block of the sequence, that in any case has no influence on the transmission rate if the number of transmitted blocks is large).

2.8 Problems

Problem 1. A binary baseband communication system makes use of the two signals

$$\begin{aligned} s_0(t) &= \begin{cases} At/T & t \in [0, T] \\ 0 & \text{elsewhere} \end{cases} \\ s_1(t) &= \begin{cases} A - At/T & t \in [0, T] \\ 0 & \text{elsewhere} \end{cases} \end{aligned}$$

The signals are transmitted with equal probability over a (real) AWGN channel with power spectral density $N_0/2$.

Determine the optimal detector that minimizes the probability of symbol error and represent its block diagram. Also, compute the symbol error probability as a function of the SNR (defined as usual by the average symbol energy over N_0).

Problem 2. Define $p_T(t)$ as

$$p_T(t) = \begin{cases} \frac{1}{\sqrt{T}} & 0 \leq t \leq T \\ 0 & \text{elsewhere} \end{cases}$$

and consider the signal set

$$\begin{aligned} s_1(t) &= \sqrt{2\mathcal{E}} \cos(2\pi f_1 t) p_T(t) \\ s_2(t) &= \sqrt{2\mathcal{E}} \cos(2\pi f_2 t) p_T(t) \\ s_3(t) &= -\sqrt{2\mathcal{E}} \cos(2\pi f_1 t) p_T(t) \\ s_4(t) &= -\sqrt{2\mathcal{E}} \cos(2\pi f_2 t) p_T(t) \end{aligned}$$

where $f_1 = k/T$ and $f_2 = (k+1)/T$, and k is an integer. Consider transmission over a (real) AWGN channel. The received signal is $y(t) = x(t) + n(t)$ where $x(t)$ can be one of the signals $s_1(t), s_2(t), s_3(t), s_4(t)$, with uniform probability, and where $n(t)$ is a white Gaussian noise with power spectral density $N_0/2$.

1) Find the Maximum-Likelihood detector structure that employs the minimum possible number of correlators, and represent its block diagram.

2) Find the symbol error probability as a function of E_b/N_0 for ML detection (either an exact expression or an upper bound. In the latter case, make the bound as tight as possible).

Problem 3. Consider the signal constellation defined by the points

$$\begin{aligned} s_m &= \sqrt{\mathcal{E}_1} \exp(j(\pi(m-1)/2 + \pi/4)) \quad \text{for } m = 1, \dots, 4 \\ s_m &= \sqrt{\mathcal{E}_2} \exp(j\pi(m-5)/2) \quad \text{for } m = 5, \dots, 8 \end{aligned}$$

1) Find the average energy per symbol E_s (symbols are equiprobable) as a function of $\mathcal{E}_1$ and $\mathcal{E}_2$.

2) For fixed E_s , find the ratio $\rho = \mathcal{E}_2/\mathcal{E}_1$ that maximizes the minimum squared Euclidean distance of the constellation (assume $\mathcal{E}_2 \geq \mathcal{E}_1$, i.e., consider the interval $\rho \geq 1$).

3) For the optimal ratio ρ found, determine an upperbound to the symbol error probability as a function of E_b/N_0 .

Problem 4. Compute the symbol error probability of a 2-FSK system with coherent detection, $\Delta T = 0.7$ over an AWGN channel with $E_s/N_0 = 10$ dB. *Note: Δ denotes the spacing between the two frequencies of the 2-FSK modulation and T denotes the symbol interval.*

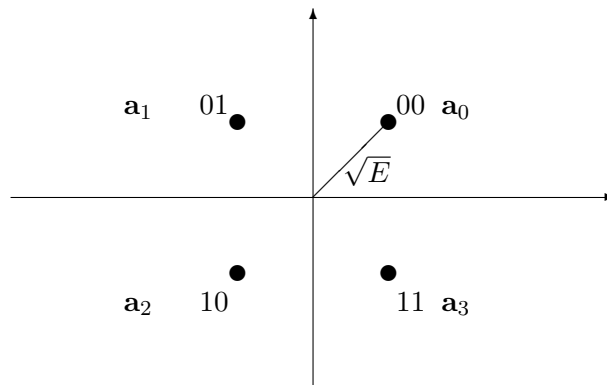
Problem 5. Repeat the previous problem in the case of non-coherent detection.

Problem 6. A digital modulator gets as input a sequence of independent and uniformly distributed bits and produces symbols a labeled by pairs of bits (b_1, b_2) according to the labeling rule

$b_1 b_2$	a
00	$-3\frac{d}{2}$
01	$-\frac{d}{2}$
11	$\frac{d}{2}$
10	$3\frac{d}{2}$

The received signal output by a (real) AWGN channel is given by $y = a + n$, where $n \sim \mathcal{N}(0, N_0/2)$. The receiver is based on the optimal decision rule that minimizes the symbol error probability. Compute the average bit-error probability.

Problem 7. In a 4-PSK modulator, each symbol is labeled by the pair of bits $(b_1 b_2)$ according to the following labeling rule:



The bits b_1 e b_2 are statistically independent with probability distribution

$$\begin{cases} P(b_i = 0) &= \varepsilon \\ P(b_i = 1) &= 1 - \varepsilon \end{cases} \quad i = 1, 2$$

(with $\varepsilon < 0.5$) The signal is modulated and transmitted over an AWGN channel with power spectral density N_0 .

Determine the decision regions of the MAP detector (i.e., the detector that minimizes the symbol-error probability), and determine the decision regions of a detector that minimizes the bit-error probability.

Problem 8. The constellation with points

$$\begin{aligned}
 s_1 &= L/2 \\
 s_2 &= -L/2 \\
 s_3 &= j\sqrt{3}L/2 \\
 s_4 &= -j\sqrt{3}L/2 \\
 s_5 &= L + j\sqrt{3}L/2 \\
 s_6 &= L - j\sqrt{3}L/2 \\
 s_7 &= -L + j\sqrt{3}L/2 \\
 s_8 &= -L - j\sqrt{3}L/2
 \end{aligned}$$

is used for transmission over an AWGN channel (note: L is a scaling factor that determines the average energy per symbol).

1) Find an upper bound on the symbol error probability for this constellation as a function of the ratio E_b/N_0 , where E_b is the average energy per bit and N_0 is the noise power spectral density.

2) Find the asymptotic gain (expressed in dB) of the above constellation with respect to an 8PSK constellation.

Problem 9. A signal set of M orthogonal signals is constructed by varying the position of a rectangular impulse. The signals are

$$s_m(t) = p(t - mT/M) \quad m = 0, 1, 2, \dots, M - 1$$

where

$$p(t) = \begin{cases} A & 0 \leq t \leq T/M \\ 0 & \text{elsewhere} \end{cases}$$

Determine the peak power level A^2 as a function of M for which this system has the same symbol error probability of an orthogonal M -FSK (with non-coherent detection).

Problem 10. Consider the on-off modulation with signals $s_0(t) = 0$ and $s_1(t) = \sqrt{2E_s}p_T(t)$, where $p_T(t)$ is a unit-energy rectangular pulse of support $[0, T]$. Find the ML non-coherent decision rule. For this rule, assuming equiprobable symbols, give an expression for the average error probability.

Problem 11. Find an exact expression of the symbol error probability for differential BPSK (non-coherent detection is performed over blocks of length $N = 2$ of adjacent symbols).

Problem 12. Compute the union bound on the symbol error probability of differential BPSK with non-coherent block detection performed on blocks of length $N = 3$ of adjacent symbols.

Problem 13. A 4-PSK modulation is used for transmission over AWGN. Due to a phase synchronization error in the receiver, the discrete-time signal at the input of the detector is given by

$$y = \sqrt{E_s}e^{j\theta}x + z$$

where z is a complex Gaussian circularly-symmetric noise with mean zero and variance N_0 , x is a symbol of the unit-energy 4-PSK constellation and θ is the phase error, assumed to be unknown and deterministic.

Compute an upper bound to the symbol error probability as a function of θ and E_s/N_0 assuming that the detector uses the same decision regions of the ML detector that ignores the phase error (i.e., that assumes $\theta = 0$).

Problem 14. Gray labeling is a binary labeling of the constellation points of a signal set such that the label of any two signals at minimum distance differ by only one bit. Consider a 8PSK constellation and find a Gray labeling. Sketch the constellation with the associated binary labels. Find an upper bound (as tight as possible) to the symbol error probability P_e with ML detection in AWGN, as a function of the SNR E_s/N_0 . Find an *exact* expression for the average bit-error probability P_b as a function of E_s/N_0 and verify that, with Gray labeling, we have $P_b \approx \frac{1}{3}P_e$ for large E_s/N_0 .

2.9 Appendix: the characteristic function of HQF-GRV

In this appendix we derive the expression (2.48) for the characteristic function of the HQF-GRV $\Delta = \mathbf{z}^H \mathbf{F} \mathbf{z}$, where $\mathbf{z} \sim \mathcal{N}_c(\bar{\mathbf{z}}, \mathbf{R})$.

Since $\mathbf{R}$ is Hermitian positive definite, it can be decomposed as

$$\mathbf{R} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^H$$

where $\mathbf{U}$ is unitary and its columns are the (normalized) eigenvector of $\mathbf{R}$, and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$ contains the eigenvalues of $\mathbf{R}$. Since $\lambda_i > 0$, we can write $\mathbf{\Lambda} = \mathbf{\Psi}^2$, where $\mathbf{\Psi} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$. The new Gaussian vector $\mathbf{w} = \mathbf{\Psi}^{-1} \mathbf{U}^H \mathbf{z}$ has i.i.d. unit covariance components, in fact

$$\begin{aligned} E[(\mathbf{w} - \bar{\mathbf{w}})(\mathbf{w} - \bar{\mathbf{w}})^H] &= \mathbf{\Psi}^{-1} \mathbf{U}^H \mathbf{R} \mathbf{U} \mathbf{\Psi}^{-1} \\ &= \mathbf{\Psi}^{-1} \mathbf{U}^H \mathbf{U} \mathbf{\Lambda} \mathbf{U}^H \mathbf{U} \mathbf{\Psi}^{-1} \\ &= \mathbf{I} \end{aligned}$$

We have the inverse relation $\mathbf{z} = \mathbf{U} \mathbf{\Psi} \mathbf{w}$, so that

$$\Delta = \mathbf{w}^H \mathbf{\Psi} \mathbf{U}^H \mathbf{F} \mathbf{U} \mathbf{\Psi} \mathbf{w} = \mathbf{w}^H \mathbf{T} \mathbf{w}$$

The Hermitian matrix $\mathbf{T} = \mathbf{\Psi} \mathbf{U}^H \mathbf{F} \mathbf{U} \mathbf{\Psi}$ can also be written as

$$\mathbf{T} = \mathbf{S} \mathbf{\Phi} \mathbf{S}^H$$

where $\mathbf{S}$ is unitary and $\mathbf{\Phi} = \text{diag}(\phi_1, \dots, \phi_n)$. We define the new Gaussian vector $\mathbf{v} = \mathbf{S}^H \mathbf{w}$ (notice that $\mathbf{v}$ has also covariance $\mathbf{I}$ since $\mathbf{S}$ is unitary). Finally, the quadratic form can be diagonalized and written as the sum of n independent terms as follows:

$$\begin{aligned} \Delta &= \mathbf{w}^H \mathbf{T} \mathbf{w} \\ &= \mathbf{w}^H \mathbf{S} \mathbf{\Phi} \mathbf{S}^H \mathbf{w} \\ &= \mathbf{v}^H \mathbf{\Phi} \mathbf{v} \\ &= \sum_{i=1}^n \phi_i |v_i|^2 \end{aligned}$$

The characteristic function $\Phi_\Delta(s)$ is obtained in product form as

$$\begin{aligned}\Phi_\Delta(s) &= E[e^{-s\Delta}] \\ &= E\left[\exp\left(-s\sum_{i=1}^n\phi_i|v_i|^2\right)\right] \\ &= \prod_{i=1}^n\Phi_{|v_i|^2}(s)\end{aligned}$$

where we define

$$\Phi_{|v_i|^2}(s) = E[\exp(-s\phi_i|v_i|^2)]$$

Since $v_i \sim \mathcal{N}_\mathbb{C}(\bar{v}_i, 1)$ and it is circularly symmetric, we have

$$|v_i|^2 = a_i^2 + b_i^2$$

where $a_i \sim \mathcal{N}(\bar{a}_i, 1/2)$ and $b_i \sim \mathcal{N}(\bar{b}_i, 1/2)$ are independent, and where $\bar{v}_i = \bar{a}_i + j\bar{b}_i$. Then, in order to complete our calculation it is sufficient to evaluate the characteristic function of the square of a real Gaussian random variable x with variance $1/2$ and mean $\bar{x}$. We have

$$\begin{aligned}f_{x^2}(s) &= E[\exp(-s\phi x^2)] \\ &= \frac{\exp\left(-s\frac{\phi}{1+s\phi}\bar{x}^2\right)}{\sqrt{1+s\phi}}\end{aligned}\tag{2.58}$$

Then, by using the above result we obtain

$$\begin{aligned}\Phi_{|v_i|^2}(s) &= f_{a_i^2}(s)f_{b_i^2}(s) \\ &= \frac{\exp\left(-s\frac{\phi_i}{1+s\phi_i}|\bar{v}_i|^2\right)}{1+s\phi_i}\end{aligned}\tag{2.59}$$

The desired result is given by

$$\Phi_\Delta(s) = \frac{\exp\left(-s\sum_{i=1}^n\frac{\phi_i}{1+s\phi_i}|\bar{v}_i|^2\right)}{\prod_{i=1}^n(1+s\phi_i)}\tag{2.60}$$

Finally, we re-write the above characteristic function directly in terms of the matrices $\mathbf{R}$ and $\mathbf{F}$. Notice that

$$\begin{aligned}\prod_{i=1}^n(1+s\phi_i) &= \det(\mathbf{I} + s\mathbf{T}) \\ &= \det(\mathbf{U}\Psi(\mathbf{I} + s\mathbf{T})\Psi^{-1}\mathbf{U}^H) \\ &= \det(\mathbf{I} + s\mathbf{R}\mathbf{F})\end{aligned}$$

In fact, $\mathbf{U}\Psi\mathbf{T}\Psi^{-1}\mathbf{U}^H = \mathbf{U}\Lambda\mathbf{U}^H\mathbf{F} = \mathbf{R}\mathbf{F}$. Also, we can write

$$\sum_{i=1}^n\frac{\phi_i}{1+s\phi_i}|\bar{v}_i|^2 = \bar{\mathbf{v}}^H\Phi(\mathbf{I} + s\Phi)^{-1}\bar{\mathbf{v}}$$

Recalling that $\Phi = \mathbf{S}^H\Psi\mathbf{U}^H\mathbf{F}\mathbf{U}\Psi\mathbf{S}$, and that $\bar{\mathbf{v}} = \mathbf{S}^H\Psi^{-1}\mathbf{U}^H\bar{\mathbf{z}}$, we obtain

$$\bar{\mathbf{v}}^H\Phi(\mathbf{I} + s\Phi)^{-1}\bar{\mathbf{v}} = \bar{\mathbf{z}}^H\mathbf{F}(\mathbf{I} + s\mathbf{R}\mathbf{F})^{-1}\bar{\mathbf{z}}$$

so that (2.48) follows.

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